

Topological Cross-Domain Synthesis of the Spherical Substrate Corpus

Information science, state-and-event calculus,
topology, local spherical support, and the computer
graphics boundary

Prepared from seven user-supplied PDFs

222 source pages

No external performance validation

Deliverables

Illustrated report, operator ontology, claim
matrix, schemas, query API, reference prototype,
source maps, and original documents

14 August 2026

Superseding the chemistry-first reading

The earlier package over-weighted chemical and biological language. This report replaces that reading. The source corpus is treated primarily as a topological and information-architectural design record whose mature center is a queryable state-and-event substrate. Chemical, biological, optical, artistic, political, and hardware passages are retained only where they instantiate, motivate, or test the shared operator vocabulary.

THE DECISIVE SOURCE RULE

Do not judge the archive as a proposal about rasterization, raymarching, voxel storage, or a conventional frame loop. Those are downstream projection tools. The authoritative object is an ontology of directly queryable relations, events, phases, identities, and transitions. [S1 p. 1]

This report remains intentionally internal to the supplied corpus. It does not externally verify performance, historical anecdotes, biological mechanisms, hardware efficiency, or topological physics. Where the report introduces information-science language such as event sourcing, materialized views, schema, provenance, or query planning, it labels that language as a cross-domain correspondence rather than a verbatim source claim.

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1. Executive correction and findings

The corrected reading is not "hidden chemistry." It is a cross-domain corpus attempting to describe a common computational substrate beneath geometry, sensing, graphics, simulation, identity, memory, and optional photonic hardware. Its strongest form is a spherical-support event calculus: a finite grammar generates a continuous state manifold; local radial-angular supports select relevance; compatibility determines whether co-located sectors can interact; explicit relation surfaces define events; typed transitions update state while preserving invariants and lineage; images are optional outputs. [S1 pp. 1, 5-7]

The decisive interpretive shift

The corpus is not primarily a chemistry decoder or a renderer proposal.

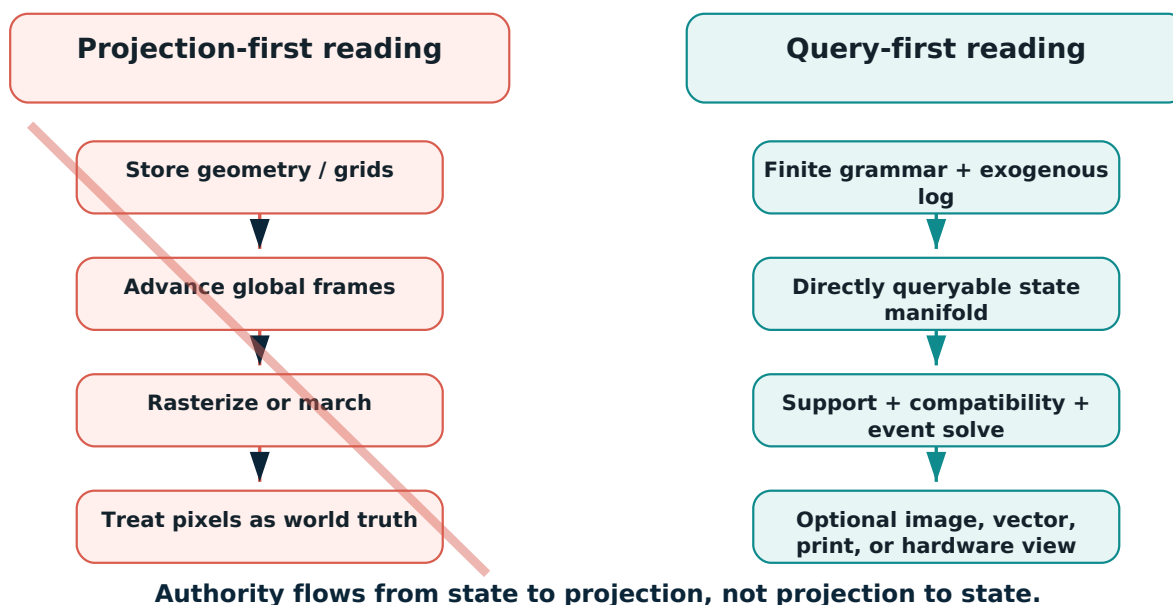


Figure 1. The corpus changes authority from projection-first to query-first.

Normalized from S1 pp. 1, 5-10 and S5 pp. 1, 9.

01

Core object

A state manifold and relation algebra, not a framebuffer, mesh inventory, or global voxel grid.

02

Topological role

Sheets, parity, gluing, overlap, and pinch points organize coupling and transition semantics.

03

Graphics role

Raster, ray, vector, display, and print pipelines are materialized views downstream of a query result.

04

Information role

Grammar, exogenous log, lineage, branch, address, and compatibility form a compact data architecture.

05

Prototype role

Equation World Zero tests six headless queries over a deliberately restricted relation family.

06

Hardware role

Hollowlens-0 is an optional bounded optofluidic implementation, not proof of the whole ontology.

What survives the full corpus

- **Retain:** direct state/event queries, local spherical supports, relation guards, compatibility, phase sheets, lineage, explicit transitions, and novelty-aware logging.
- **Translate:** double vacuum into uncoupled sectors; hourglass into routing; Klein bottle into an explicit gluing/orientation map; matrix-in-glass into a calibrated transfer matrix.
- **Demote:** claims that a geometric metaphor automatically supplies physical law, computational complexity, security, medical control, or perfect hardware behavior.
- **Reject:** universal $O(1)$, zero memory, zero latency, zero heat, literal topological immunity, and direct biological control as established consequences.

BOUNDED FEASIBILITY VERDICT

The corpus is internally coherent and prototype-worthy when restricted to controlled relation families, headless state/event queries, explicit compatibility, finite grammar depth, and testable lineage. As a universal replacement for graphics, simulation, memory, cognition, and semantics at once, it remains unproven. [S1 pp. 1, 7-12]

2. Method, source authority, and evidence boundary

The report uses a layered authority model. S1 supplies the mature interpretive rule. Earlier files contribute generative vocabulary, examples, and constraints. S6 supplies the most disciplined physical translation. S7 is treated as a graphics projection adapter. Where sources conflict, later normalization controls unless the report explicitly preserves the conflict as an unresolved design choice.

Seven-document source architecture

Generative prose is normalized by later interpretive and engineering discipline.

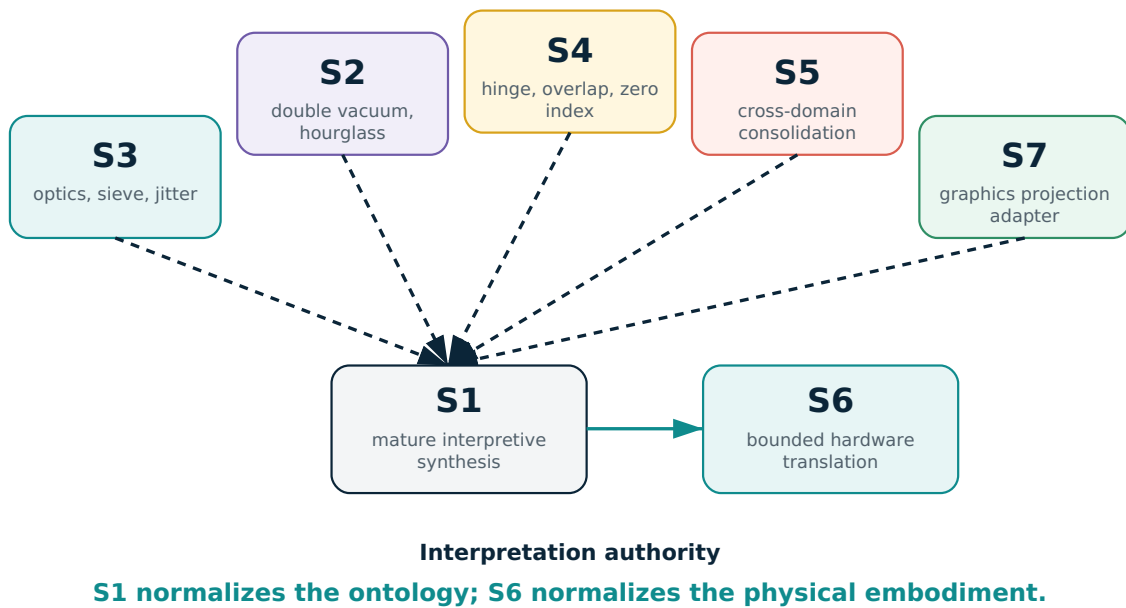


Figure 2. Source authority is asymmetric: S1 normalizes ontology; S6 normalizes physical embodiment.

Derived from the source-role map in S1 pp. 2-5 and S6 p. 11.

Evidence labels

Label	Meaning in this report
canonical	Explicitly supported by the mature interpretive synthesis or bounded engineering source.
normalized operator	A recurring source motif translated into a testable computational role.
decoder inference	An information-science or graphics correspondence added by this report and marked as such.
bounded hypothesis	A source proposal retained only as an experiment with exit criteria.
rejected totalizing claim	A claim the source itself later narrows or the bounded report explicitly rejects.

NO SILENT REPAIR

Equations in this report are either reproduced from the mature formal core, presented as normalized notation, or explicitly labeled as a prototype restriction. Missing source equations are not silently invented. The supplied texts are design records; this package does not convert metaphor into empirical fact.

Scope exclusions

The technical core excludes personal, political, punitive, medical, and bodily-control narratives as proof of a computational architecture. It also excludes chemical structures as a primary decoding target. Those passages remain in the source extracts for provenance, but they do not govern the cross-domain model.

[S1 pp. 11-12; S6 p. 10]

3. Corpus architecture and chronology

The archive stabilizes by correction rather than by a single clean specification. Early optical and shader language supplies apertures, cones, polar charts, and sensory filters. Middle documents introduce parity, zero-crossings, folds, and persistent anchors. Hollowland expands these into phase sheets, a quad hourglass, gluing, and direct event solving. ADDominUS consolidates identity, memory, relation algebra, and hardware prompts. The chronological synthesis then changes the reading rule, and the spherical-throughput report bounds the physical implementation.

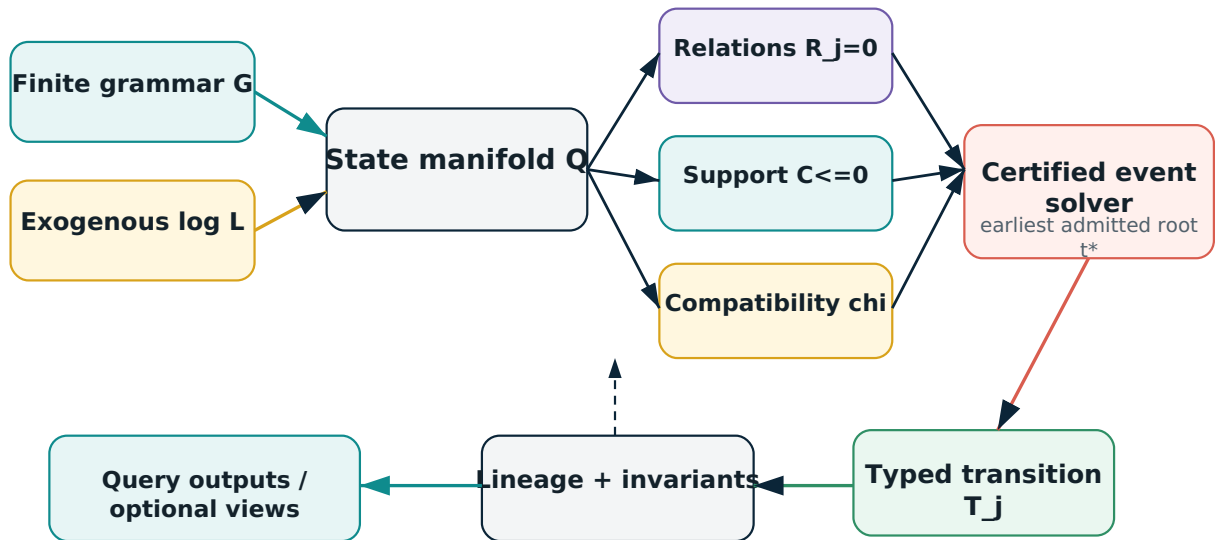
ID	Document	Role	Status
S1	Chronological Synthesis of the Spherical Substrate Line(3).pdf	Mature interpretive synthesis and feasibility discipline	canonical interpretive authority
S2	hollowland-double-vacuum(7).pdf	Generative topology and CS vocabulary: double vacuum, SDF zero, phase, parity, hourglass, eigenspace	exploratory and metaphor-heavy
S3	dmt-spirit-molecule-(leeuwenhoek-animalcules-peter-the-great-netherlands-signify-phillips-netherlands) (1).pdf	Optical/perceptual origin motifs: aperture, inversion, polar LUTs, sensory sieves, jitter, reflection	exploratory cross-domain narrative
S4	21BenBurgersStrikesBackTelNetNiet.pdf	Zero-index, hinge, phi/D-fold, overlap and shared-domain motifs	geometric/typographic brainstorming
S5	ADDominUS(1).pdf	Cross-domain consolidation: relation algebra, identity, memory, compatibility, hardware prompts	mixed synthesis with speculative claims
S6	Spherical_Throughput_Practical_Waveguide_Liquid_Substrate_Lensing(sqwit (super quick wit).pdf	Bounded engineering translation, B.C.E. metrics, Hollowlens-0, failure criteria	bounded implementation guide
S7	KC Vector Graphical Art System.pdf	Downstream computer-graphics projection adapter: log-polar SDF, 1-bit modulation, jitter and prepress	projection-layer proposal

Direction of correction

The important chronological movement is from display vocabulary to local sensing, and from local sensing to a general event calculus. "Sphere" ceases to mean a global replacement coordinate system and becomes a local chart for support, shelling, orientation, and radial reach. "SDF zero" ceases to mean an invitation to keep sampling and becomes the event locus. "One bit" ceases to describe the world and becomes a narrow routing bit. "Double vacuum" ceases to describe emptiness and becomes the absence of coupling. [S1 pp. 2-7]

Spherical-support event calculus

The core pipeline is declarative and headless.



Event = relation crossing AND support admission AND compatibility.

Figure 3. The mature core is a declarative event pipeline.

S1 pp. 5-7, 11-12; S6 pp. 3, 7.

4. Normalization rules: from prose motifs to operators

The source vocabulary is intentionally overloaded. The same word may refer to geometry, optics, phase, computation, identity, or metaphor. A useful report therefore cannot merely collect nouns. It must assign each noun a type, scope, input, output, and status. The following normalization table is the center of that conversion.

Source motif	Normalized operator	Disposition	Source basis
sphere / spherical	local radial-angular support, shelling, orientation, radial reach	retain and narrow	[S1 pp.1,5-7]
cone / lighthouse	analytic support or relevance operator over state/query domain	retain and normalize	[S1 pp.5-7,10; S2 pp.18-19]
SDF=0 / B=0	explicit event or transition guard surface	retain as core	[S1 p.7; S2 pp.16-19]
double vacuum	same coordinate but incompatible sector; absence of coupling	retain as core	[S1 p.6; S5 pp.32-33]
one bit	parity, route, validity, or admission bit with declared schema	retain narrowly	[S1 pp.5-6; S6 p.7]
quad hourglass	four-sector routing partition with central event locus	retain as visualization	[S1 p.6; S2 pp.23-25]
Klein bottle	quotient/gluing/orientation map; explicit ports and transfer required	demote literal hardware	[S1 p.6; S6 pp.2,5,10]
Mobius strip	optional local orientation/inversion motif	metaphorical/optional	[S3 pp.5-10]
phi	split into angular coordinate, phase law, or heuristic schedule	disambiguate	[S2 pp.14-19; S5 pp.1-3]
golden ratio	optional schedule or aesthetic motif only	reject as requirement	[S2 pp.14-19; S4 pp.10-12]
sieve	sensor transfer, filter, candidate reduction, or materialized index	retain with caution	[S3 pp.11-15; S5 pp.13-16]
jitter / wiggle	controlled sampling/dither/micro-shift operator	retain layer-specifically	[S3 pp.13-15; S7 pp.4-10]
Feynman vectors	phase-amplitude edge accumulator (source term is not formally derived)	rename for clarity	[S7 pp.1-3]
matrix-in-glass	calibrated passive or tunable optical transfer matrix	bounded translation	[S6 pp.2,4-5]
liquid substrate	packaged liquid lens/overclad/core tuning focus, n_{eff} , phase, coupling	bounded translation	[S6 pp.3-5,8-9]
zero latency / heat / memory	not accepted; include end-to-end costs and stored novelty	reject	[S6 pp.2,10]

Source motif	Normalized operator	Disposition	Source basis
universal O(1)	only horizon-independent evaluation for fixed closed expressions may be tested	reject totalizing form	[S1 p.8]
chemical/biological literalization	cross-domain analogy or unsupported application unless a separate model is specified	exclude from core	[S1 pp.11-12; S6 p.10]

Claim discipline used in the new report

Keep useful operators; separate source definition from validation.

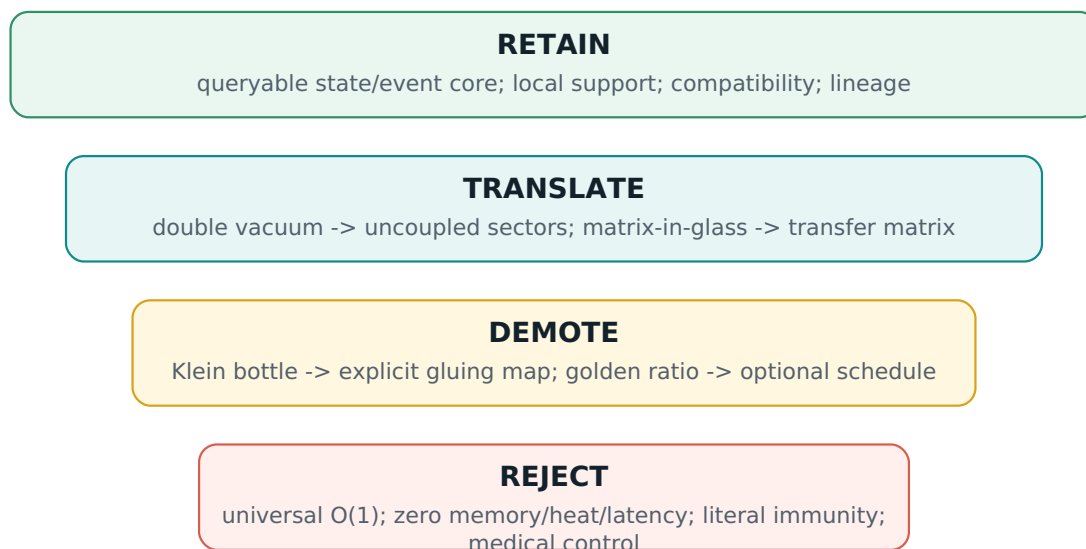


Figure 4. The report keeps useful operators while separating source definition from validation.

Disposition policy synthesized from S1 and S6.

TYPE DISCIPLINE

Optical field is governed by the optical model; **liquid interface** by fluid and surface mechanics; **guided amplitudes** by coupled-mode theory; **event declaration** by compatibility and guard logic. Topological language cannot substitute for the equations of a physical layer. [S6 pp. 2, 5]

PART II

Formal architecture of the spherical-support event calculus

The corpus becomes technically legible when its operators are typed as state, relation, support, compatibility, event, transition, identity, and projection.

5. The query-first core

The central inversion is a change in the question asked by the system. A conventional scene pipeline asks what geometry is stored, what must update this frame, and how a sensor produces an image. The equation-world asks which finite relations generate possible states, what state holds at a requested time, which event occurs next, and which sectors are allowed to couple. That is a shift from picture production to direct state semantics.

[S1 pp. 9-10]

Projection-first question

- Which surfaces and samples exist now?
- How are they projected this frame?
- Which pixels or rays are evaluated?
- How is temporal state replayed?

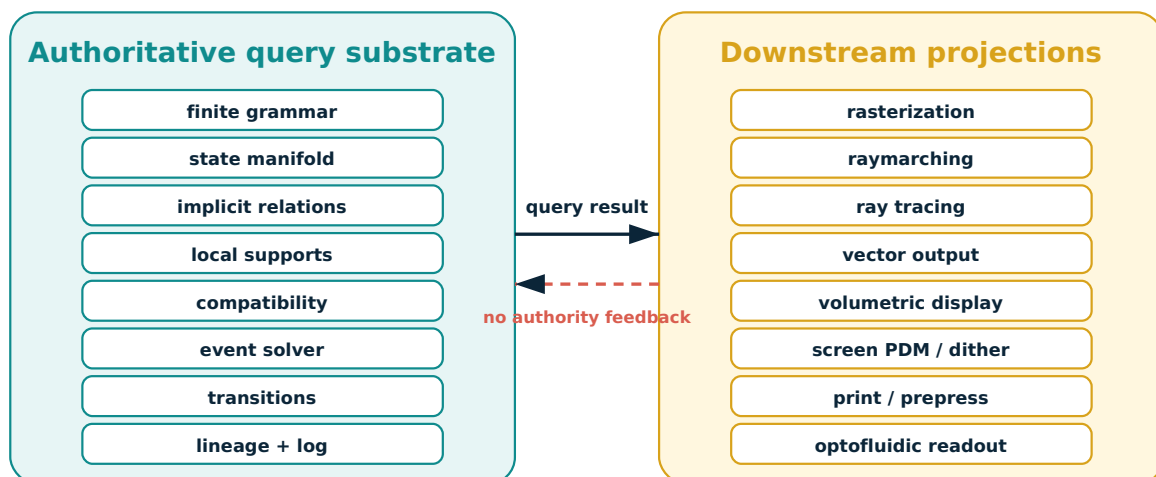
Query-first question

- Which relations define the admissible state?
- What is $q_e(t)$ at requested time t ?
- What is the earliest compatible guard crossing?
- Which identity and invariant survive the transition?

This does not ban graphics. It relocates graphics. A projection may be raster, ray-based, vector, volumetric, print, sensor, or photonic. None of those output forms is allowed to overwrite the authority of the state manifold.

Computer graphics: core versus projection

The corpus overlaps with graphics without reducing the world to drawing.



Do not mistake the projector for the state space.

Figure 5. Query substrate and projection systems are connected, but authority does not flow backward from pixels to ontology.

S1 pp. 9-10; S7 pp. 1-10.

6. State product and entity identity

The mature formal condensation treats state as a product space. Position is only one coordinate family. Time, phase, sheet, lineage address, and branch are independent dimensions of the computational state. This is the formal reason that co-location does not force interaction and that identity does not collapse into a current coordinate. [S1 pp. 5-6]

NORMALIZED PRODUCT STATE SPACE

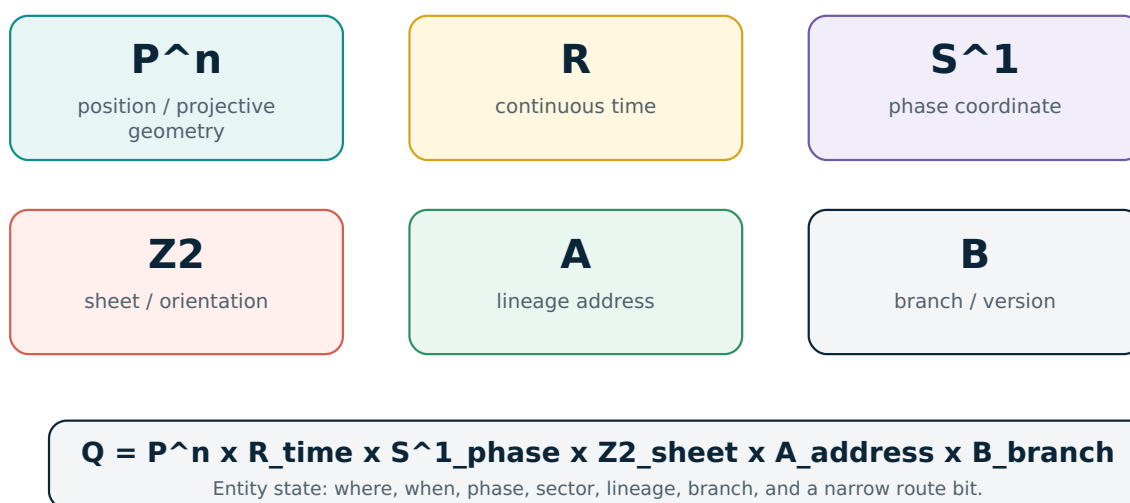
$$Q = P^n \times R_{\text{time}} \times S^1_{\text{phase}} \times Z2_{\text{sheet}} \times A_{\text{address}} \times B_{\text{branch}}$$

ENTITY STATE USED IN THIS REPORT; P_E IS A NARROW PARITY/ROUTE BIT

$$q_e(t) = (x_e(t), t, \phi_e(t), \sigma_e(t), a_e, b_e, p_e)$$

State is more than position

Normalized product space from the mature corpus reading



Images, meshes, and frames are views derived from Q.

Figure 6. State contains geometry, time, phase, sector, lineage, and branch.

S1 pp. 5-6. The explicit parity component is added as a narrow routing field from S6 p. 7.

World description

SOURCE FORMAL CORE PLUS EXPLICIT EXOGENOUS LOG

$$W = (G, R, T, I), \text{ with an external novelty log } L$$

- G: finite grammar or generative program.

- R: relation and guard family.
- T: typed transition rules.
- I: invariants that must survive designated transitions.
- L: exogenous log for shocks, interventions, or novelty not implied by the closed grammar.

Identity

The corpus repeatedly insists that identity is not exhausted by a snapshot. The normalized information-science form is a lineage address derived from seed, grammar path, and invariant history. The hash form below is a prototype representation, not a source-mandated cryptographic standard.

DECODER NORMALIZATION OF ONTOLOGICAL ADDRESS

$$a_e = H(\text{seed}_e, \text{grammar_path}_e, \text{lineage_history}_e)$$

IDENTITY INVARIANT

Person(*t*) != Snapshot(*t*) is best read as an architectural caution: current position, role, image, or record is a projection of identity, not the full identity. [S5 pp. 6-9]

7. Relation, support, compatibility, event, transition

A pure geometric intersection is insufficient. The source formal core requires three conditions: the relation or guard reaches its event surface; the state lies within a declared local support; and the compatibility predicate admits coupling. Only then is the transition invoked. [S1 pp. 5-7; S6 pp. 3, 5, 7]

RELATION OR GUARD SURFACE

$$R_j(q; \theta_j) = 0$$

LOCAL SPHERICAL, CONIC, SHELL, CAUSAL, OR POLICY SUPPORT

$$C_{\alpha}(q, t) \leq 0$$

COMPATIBILITY PREDICATE

$$\chi(e, j, t, \text{context}) \in \{0, 1\}$$

EARLIEST ADMITTED EVENT

$$t^* = \min \{ t \geq t_0 : R_j(q_e(t)) = 0 \text{ AND } C_{\alpha}(q_e(t), t) \leq 0 \text{ AND } \chi(e, j, t) = 1 \}$$

TYPED TRANSITION

$$q_e(t^*+) = T_j(q_e(t^*-), \text{context})$$

Why SDF zero matters

Earlier passages sometimes imagine "going past zero" in a continuous field as if that alone escaped rasterization. The mature synthesis corrects this: the zero set is not a magical continuation surface. It is the explicit locus where event semantics are invoked. The engine solves for a root or certified interval and then applies a transition. If it merely samples the field again and again, hidden marching has returned.

[S1 p. 7; S2 pp. 16-19; S3 pp. 30-33]

Typed event record

```
Event {  
  event_id, entity_id, relation_id, transition_id,  
  t_star, support_admitted, compatible, predicate_version,  
  guard_crossed, root_multiplicity, confidence,  
  pre_state_hash, post_state_hash,  
  lineage_parent, branch_created,  
  parity_before, parity_after,  
  invariant_report  
}
```

The event record makes the architecture auditable. A result is not merely "a hit"; it carries the support, predicate version, guard, uncertainty, lineage, and transition outcome that justified it.

8. Double vacuum as relational exclusion

Double vacuum is the most successful cross-domain operator in the corpus. Its mature meaning is not a second or stronger emptiness. It means that two states may share a spatial coordinate while remaining unavailable to one another because their phase, sheet, orientation, lineage address, mode, time window, or policy is incompatible. The second "vacuum" is the absence of coupling. [S1 p. 6; S5 pp. 32-33]

Double vacuum = same coordinate, different sector

The second vacuum is absence of coupling, not stronger emptiness.

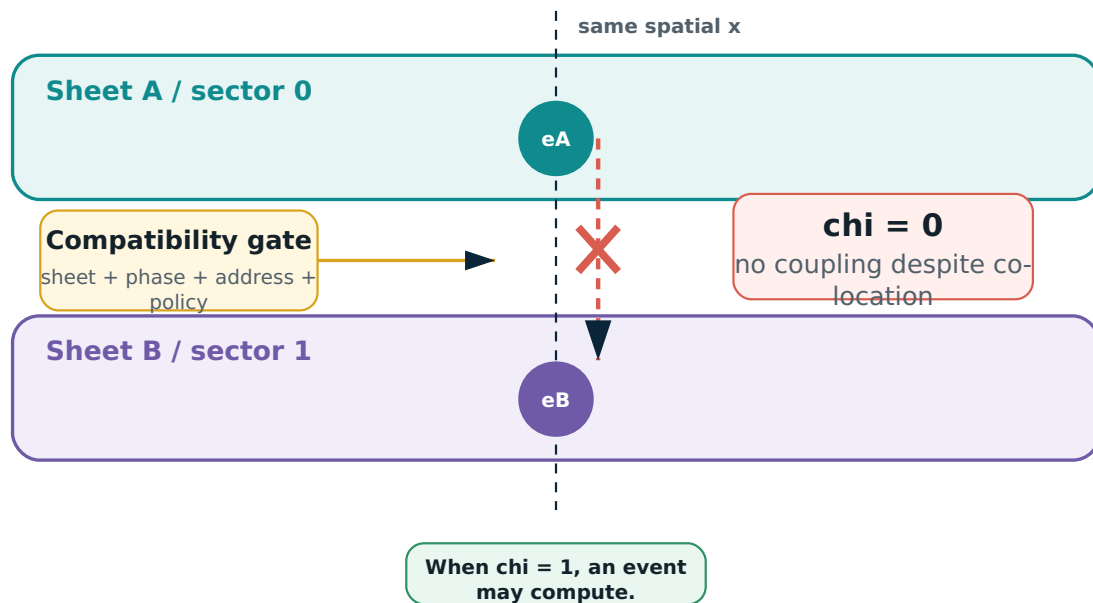


Figure 7. Co-location is not coupling. A compatibility predicate controls whether sectors can interact.

S1 p. 6; S5 pp. 32-33; S6 pp. 2, 7.

CORE AXIOM

$x_A = x_B$ does not imply $\text{can_couple}(A,B)$

NORMALIZED PREDICATE

$\text{can_couple}(A,B,\text{context}) = \text{chi}(\text{sheet}, \text{phase}, \text{address}, \text{mode}, \text{time}, \text{policy})$

Information architecture reading

- **Namespace separation:** identical keys or coordinates can exist under different sheet or branch contexts.
- **Access control:** compatibility is a declared join and permission predicate.
- **Counterfactual isolation:** branches can share a base state without contaminating one another.

- **Mode isolation:** in the bounded photonic translation, uncoupled optical modes or channels instantiate the same principle.

NOT A PHYSICAL SHIELD

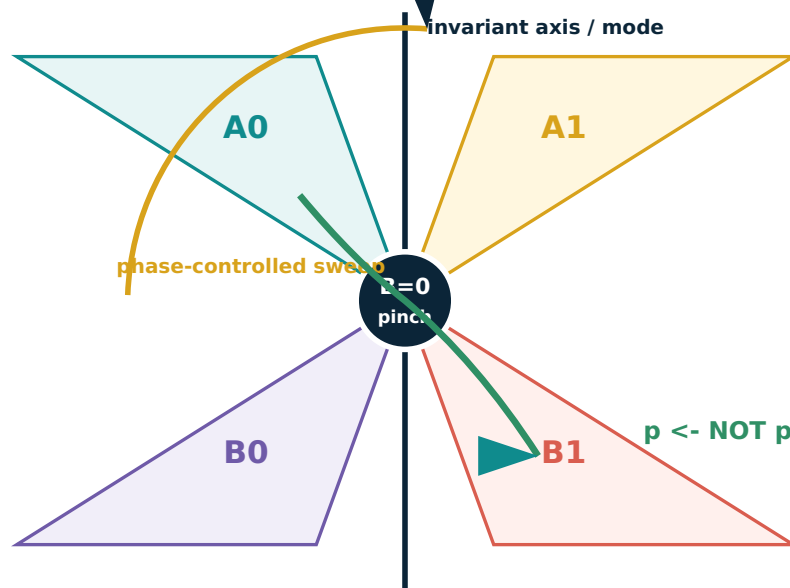
Double vacuum does not establish literal immunity, ghost-like passage, bodily control, or a new vacuum law. Those literalized claims are excluded. In hardware, the bounded translation is uncoupled or deliberately isolated modes and channels. [S6 pp. 2, 10]

9. Hourglass, parity, pinch, and gluing

The quad hourglass condenses four recurring motifs: a four-sector partition, a central pinch or rank-changing event locus, a vertical invariant axis, and a swept phase-controlled coupling arc. In the mature reading, it is a routing visualization and transition table. It is not by itself a proof of non-orientable physics. [S1 p. 6; S2 pp. 23-25]

Quad hourglass as a routing visualization

Four sectors, one event locus, explicit parity and transition rules.



The figure organizes routing; it is not a literal topology or physics proof.

Figure 8. The hourglass organizes four sectors around a central event guard.

Normalized from S1 p. 6 and S2 pp. 23-25.

One-bit parity

The bit has a narrow role. It may encode route A/B, orientation parity, admitted/rejected, fresh/stale, or available/unavailable, but only under a declared schema. The bit never contains the optical amplitude, full geometry, uncertainty, lineage, policy, and branch state. [S1 pp. 5-6; S6 p. 7]

POSSIBLE TRANSITION-SIDE PARITY TOGGLE

$$p_{\text{after}} = \text{NOT } p_{\text{before}}$$

Klein bottle and Mobius language

A Klein-bottle or Mobius motif is retained only as an orientation or gluing abstraction. To become executable, it needs explicit ports, an orientation map, a quotient or identification rule, and a transfer function. Without those objects, the topology remains metaphor. The bounded hardware report makes this discipline explicit.

[S6 pp. 2, 5, 10]

Operationally retained

- sheet identification
- orientation flip
- route parity
- port-to-port gluing
- branch transition

Not automatically supplied

- physical containment
- zero tunneling
- constant complexity
- security or bodily immunity
- electromagnetic or fluid law

PART III

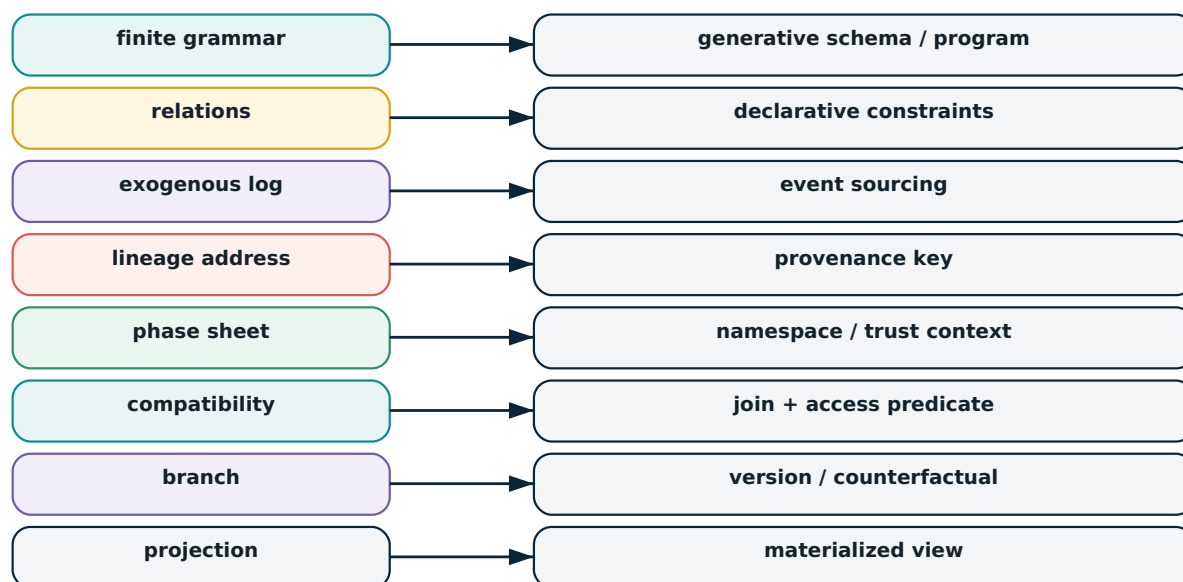
Information-science normal form

The alchemical/topological vocabulary can be organized as a data architecture: program, state schema, relation family, compatibility predicate, event log, provenance key, branch context, and materialized view.

10. Information-science interpretation

Information-science normal form

Alchemical/topological vocabulary translated into data architecture roles



The mapping is an interpretive normalization, not an external replacement of the source.

Figure 9. Source motifs mapped to information architecture roles.

The left-hand terms are source vocabulary. The right-hand terms are report-level information-science correspondences.

The mapping is useful because it makes implementation obligations visible. A "finite grammar" must become a typed generative program. A "relation" must become a declarative predicate or solvable expression. A "phase sheet" must become a namespace or context key. A "lineage address" must become a provenance identifier with split/merge rules. A "projection" must become a materialized view whose loss and staleness are understood.

Normal-form components

Corpus term	Information-science correspondence	Required explicit object
Finite grammar	Generative schema/program	Versioned rule set, input types, closure limits
State manifold	Typed state model	Schema for geometry, time, phase, sheet, address, branch, parity
Relation algebra	Declarative constraint system	Expression family, parameters, numerical solver contract
Compatibility	Join and access predicate	Boolean plus reason codes and predicate version
Event	Event-sourced fact	Timestamp/interval, guard, confidence, before/after state hashes
Transition	State-machine command	Typed mutation, invariant checks, branch policy
Lineage	Provenance graph	Parent/child edges, split/merge/reconciliation rules

Corpus term	Information-science correspondence	Required explicit object
Exogenous log	Append-only novelty stream	Source, timestamp, authority, replay semantics
Projection	Materialized view	Query, transform, resolution, error, refresh policy

DECODER INFERENCE

Terms such as event sourcing, materialized view, provenance graph, and query planner are not consistently used in the source PDFs. They are the report's information-science normal form for making the source operators implementable.

11. Memory, lineage, branches, and event sourcing

The corpus proposes that closed dynamics need not be stored frame by frame. A seed and finite grammar regenerate predictable structure; an exogenous log stores only irreducible novelty; branch history and lineage store the information needed to reconstruct splits, merges, and interventions. The mature report treats this as a bounded hypothesis, not a zero-memory theorem. [S1 pp. 7-8; S5 pp. 10-11]

BOUNDED NOVELTY-PROPORTIONAL MEMORY HYPOTHESIS

$M_{\text{required}} \sim \text{seed} + \text{grammar} + \text{exogenous_log} + \text{necessary_branch_history}$

What can be regenerated

- Closed trajectories defined by fixed expressions.
- Implicit geometry defined by finite coefficients.
- Deterministic transitions whose inputs and branch context are known.
- Derived images, meshes, and reports as materialized views.

What still accumulates

- External interventions and sensor surprises.
- Ambiguous or non-deterministic branch choices.
- Predicate and schema revisions.
- Identity split/merge reconciliation.
- Numerical certificates, uncertainty, and calibration state.

This is an event-sourced design pattern: state may be recomputed from compact generative structure plus an authoritative log. The advantage exists only when replay, solving, and branch resolution remain cheaper than storing or materializing the avoided structure. [S1 p. 8]

Lineage rules

```
lineage_address = H(seed, grammar_path, ordered_parent_events)

split(parent, labels):
    create child branches with parent_address and unique branch labels

merge(parents, policy):
    require explicit reconciliation policy
    record all parent addresses and invariant conflicts

reconstruct(address, t):
    load seed + grammar version
    apply admitted exogenous events up to t
    verify invariant and predicate versions
```

12. Query API and execution model

The core is best judged by the questions it can answer without reconstructing a conventional scene pipeline. The source experiment program names state-at-time, next-event, events in support, phase coupling, transition routing, and identity reconstruction. This package adds reason-coded compatibility and explicit projection as interface operations. [S1 pp. 11-12]

Query	Inputs	Dominant operation	Output	Failure modes	Source
state_at	entity, time	Closed trajectory/ grammar expression	State tuple $q_e(t)$	Expression size; branch history; conditioning	[S1 pp.7-8,10-12]
next_event	entity, t_0 , relation family	Compatibility filter plus certified root solver	Earliest admitted event	Event density; multiple roots; tangency; degeneracy	[S1 pp.7-8,11-12]
events_in_s upport	support, interval, policy	Analytic support pruning then relation evaluation	Ordered event set	Poor support/ compatibility turns into broad scan	[S1 pp.6-8,11]
can_couple	entity A, entity B, context	Sheet/phase/address/ mode/policy predicate	Boolean plus reason code	Predicate design; semantic grounding	[S1 p.6; S5 pp.32-33; S6 p.7]
transition	event, pre-state, context	Typed transition rule	Post-state, parity route, invariant check	Ambiguous or non- confluent rules	[S1 pp.6-7; S6 p.7]
reconstruct _identity	seed, grammar path, log, branch	Lineage replay/reduction	Identity/provenance record	Split/merge collision; log incompleteness	[S1 pp.8,11-12; S5 pp.7-11]
project_vie w	query result, camera/display policy	Downstream renderer or vector adapter	Image/vector/print output	Projection errors do not alter authoritative state	[S1 pp.9-10; S7 pp.1-10]
measure_thr oughput	event log, time window, error boundary	Count verified B.C.E. events only	Theta_sph and benchmark vector	Miss/false events; energy boundary; calibration	[S6 p.6]

Reference execution plan

```

function NEXT_EVENT(entity e, time t0, relation_family J):
    candidates = []
    for relation j in J:
        if not policy_can_apply(e, j): continue
        roots = certified_roots(R_j(q_e(t)) = 0, t >= t0)
        for r in roots:
            state = state_at(e, r)
            if support_contains(C_j, state) and chi(e, j, r) == 1:
                candidates.append(certified_event(j, r, state))
    return earliest_ordered(candidates)

function APPLY_TRANSITION(event ev):
    verify guard, predicate version, confidence, and pre-state hash
    post = T_ev(pre_state, context)
    verify invariants
    append event and lineage edges
    return post

```

Query planner obligations

- Choose a restricted relation family with known or certified root methods.
- Use support and compatibility to prune before expensive solving.
- Return intervals or uncertainty when a unique exact root is not certified.
- Make branch precedence and event ordering explicit.
- Never treat absence of a found root as proof of absence without solver bounds.

PART IV

Computer graphics and sensing boundary

The corpus overlaps deeply with graphics, but its mature claim is query-versus-projection, not sphere-versus-grid.

13. Computer graphics boundary

Rasterization, raymarching, and ray tracing answer projection and transport questions. The proposed substrate answers state, event, compatibility, identity, and reachability questions. The two layers overlap because implicit geometry, cones, local charts, and visibility can feed images. They remain distinct because the image is an output of a query, not the ontological store. [S1 pp. 9-10]

System	Primary question	Overlap with the corpus	Why it is not the core
Rasterization	How do stored surfaces project into pixels now?	Can emit a view from queried geometry/state.	Pixels and frames are not authoritative world state.
Raymarching	What does repeated field sampling along a projected direction reveal?	Uses SDFs and directional queries.	Repeated sampling is explicitly not the mature event-solving core.
Ray tracing	Which transport/visibility paths connect endpoints?	Cones and supports may define candidate paths.	The core cone also represents causal, semantic, collision, memory, and grammar support.
Vector graphics	Which analytic paths should be stroked/filled?	Implicit boundaries can become vector contours.	Output paths do not define phase, lineage, branch, or event authority.
Proposed substrate	What state holds, what event occurs next, and which sectors can couple?	Can project to any of the above.	Direct relation algebra and event calculus are constitutive.

The correct comparison is therefore not "spherical graphics versus Cartesian graphics." It is "query semantics versus projection semantics." Local polar or spherical charts may improve a sensor or output adapter without requiring global remeshing of the state model.

14. KC Vector as a downstream projection adapter

The KC Vector document is most useful as a projection-layer case study. It starts with a local log-polar address, evaluates implicit SDF/CSG boundaries, accumulates phase and amplitude near an edge, and quantizes the result for display or print. This is compatible with the mature corpus only when KC output is treated as a view over the query substrate, not as the substrate itself. [S7 pp. 1-10]

KC Vector system as a projection adapter

A downstream path from implicit query results to screen or print

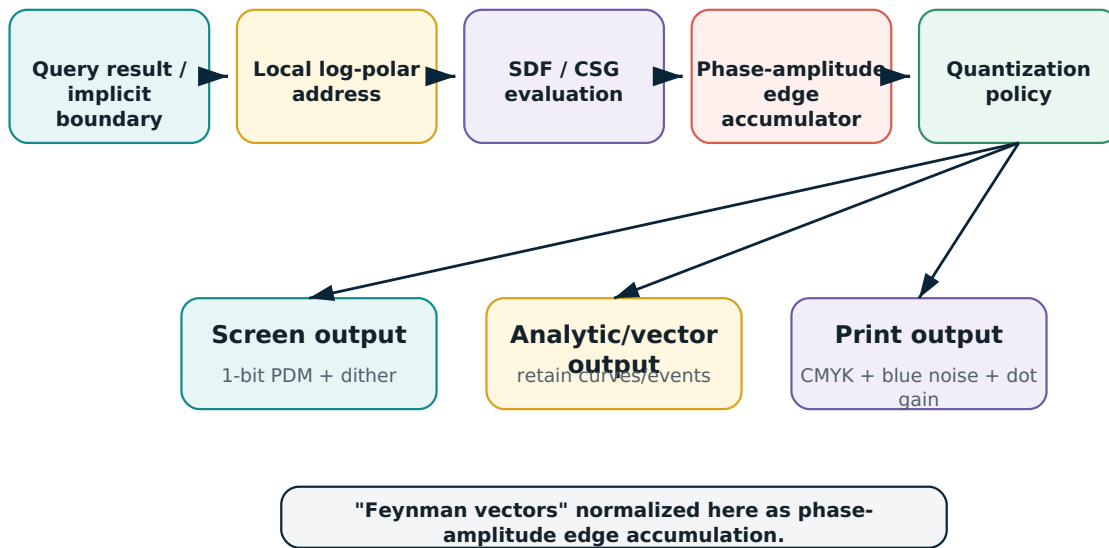


Figure 10. KC Vector is normalized as a projection adapter.

S7 pp. 1-10. "Feynman vectors" are renamed phase-amplitude edge accumulation because the source does not supply a formal path-integral derivation.

Projection stages

LOCAL LOG-POLAR ADDRESS TRANSFORM FROM S7

$$\rho = \ln(\sqrt{x^2 + y^2}), \theta = \text{atan2}(y, x)$$

SOURCE SDF/CSG OPERATORS

$$\text{union} = \min(f_1, f_2), \text{intersection} = \max(f_1, f_2), \text{subtraction} = \max(f_1, -f_2)$$

$$A(p) = \sum_k a_k \exp(i * \phi_k)$$

1. **Query result:** obtain an implicit boundary or relation slice from the authoritative substrate.
2. **Local address:** map the output domain to log-polar coordinates where radial scaling becomes translation in rho.
3. **Boundary evaluation:** evaluate SDF/CSG and its gradient near the requested contour.
4. **Phase-amplitude accumulation:** combine local edge contributions. This is the disciplined replacement for the source's under-specified "Feynman vector" term.
5. **Quantization:** choose a screen, vector, or print policy.

Screen and print paths

- **Screen:** 1-bit pulse-density modulation and anisotropic dithering distribute tone temporally and spatially.
- **Vector:** contour extraction preserves analytic edges for scalable output.
- **Print:** CMYK mapping, log-polar blue-noise screening, SDF edge locking, and dot-gain compensation create a static 1-bit prepress representation.

PROJECTION-LAYER CAVEAT

These display and prepress mechanisms do not prove that the world model is 1-bit, log-polar, or quantum. They show how a rich query result can be emitted through a highly constrained output channel.

15. Apertures, sieves, jitter, and local charts

The optical/perceptual source contributes a productive analogy: every sensor is a local transfer function with an aperture, field of view, angular/radial support, spectral response, sampling mosaic, and uncertainty. The source uses microscopes, camera obscura, compound eyes, retinal mosaics, whiskers, sonar, UV response, and artistic inversion to explore that idea. The mature architecture keeps the local support and filter, not the literal equivalence of all sensors. [S3 pp. 1-20, 24-33]

Sieve

A sieve can mean a physical receptor mosaic, a spatial index, a mode filter, a support predicate, or a quantization grid. The report therefore requires a layer tag whenever the term is used. "Bypass the sieve" can mean avoiding unnecessary materialization or broad-phase scanning, but it cannot mean that all filtering disappears. Compatibility and support are themselves filters.

Jitter

Jitter is retained as a controlled sampling or dithering operator. It can shift an aperture, sensor, or quantization pattern to fill blind spots, estimate uncertainty, or decorrelate aliasing. It does not restore a mathematically continuous world from finite samples without an error model.

LAYER-SPECIFIC JITTER MODEL

$$q_sample(t) = q_nominal(t) + \epsilon(t), \text{ with declared spectrum and bound}$$

Local spherical chart

The strongest spherical claim is local: a sensor, coupler, agent, or query has radial reach, angular support, orientation, shell structure, and uncertainty. The world may remain hybrid. This avoids both raster-everything and polar-everything. [S1 pp. 1, 6; S6 p. 3]

PART V

Prototype, measurement, and research discipline

A useful first build is deliberately small, headless, typed, and falsifiable.

16. Equation World Zero

Equation World Zero is the corpus's proper first experiment. It uses a restricted state space, simple analytic trajectories, a bounded relation family, explicit sheets and parity, finite grammar depth, lineage-based identity, and symbolic query outputs. It does not begin with a renderer. [S1 pp. 7, 11-12]

Equation World Zero

Restricted, headless, falsifiable prototype

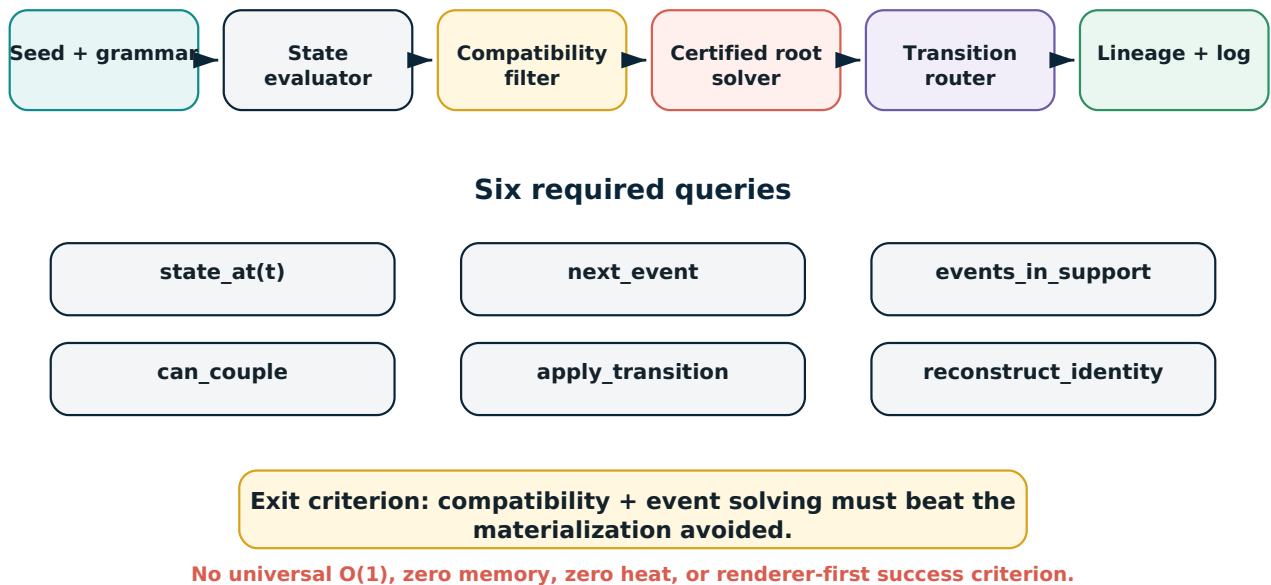


Figure 11. Equation World Zero tests six required queries and a single decisive exit criterion.

S1 pp. 7, 11-12.

Reference restriction in this package

- 2D constant-velocity trajectories plus continuous time and linear phase schedule.
- Line and circle implicit relations with analytic roots.
- Annular-sector local support.
- Sheet, phase, and invariant-tag compatibility.
- Typed transition with optional sheet change, parity toggle, branch creation, and state update.
- Hash-based lineage address and append-only event log.

WHAT THE CODE DEMONSTRATES

The included Python prototype demonstrates closed-form state_at, analytic line/circle event roots, compatibility rejection for co-located sheets, parity routing, lineage update, and an auditable event record. Four unit tests pass. It is not a renderer, topology proof, general physics engine, or universal solver.

Six required core queries

state_at(t)

next_event(t0)

events_in_support

can_couple

apply_transition

reconstruct_identity

17. Worked event-routing example

The package includes a deterministic example that turns the prose operators into an auditable trace. An entity begins at $(-2, 0.25)$, moves right at unit speed, occupies sheet 0, and carries invariant tag `mass-class-A`. Relation `B0` is the line $x=0$. Its support and compatibility policy admit sheet 0. The earliest root is $t^*=2$. The transition moves the entity to sheet 1, toggles parity, creates a branch, updates the route field, and derives a new lineage address.

Pre-state

```
{
  "entity_id": "e-alpha",
  "time": 0.0,
  "position": [
    -2.0,
    0.25
  ],
  "phase": 0.0,
  "sheet": 0,
  "address":
    "0c34c34eab1a93ccece2456125c824a2",
  "branch": "root",
  "parity": 0,
  "invariant_tags": [
    "mass-class-A"
  ],
  "state_variables": {}
}
```

Post-state

```
{
  "entity_id": "e-alpha",
  "time": 2.0,
  "position": [
    0.0,
    0.25
  ],
  "phase": 0.0,
  "sheet": 1,
  "address":
    "b29237d737a4dbc18a5e4b1b895c0a52",
  "branch": "root/route-A-to-B@2",
  "parity": 1,
  "invariant_tags": [
    "mass-class-A"
  ],
  "state_variables": {
    "route": "B"
  }
}
```

Event certificate

Field	Value
event_id	8b1aa2790d0369fe7b3d4152
entity_id	e-alpha
relation_id	B0
transition_id	route-A-to-B
time	2.0
support_admitted	True
compatible	True
compatibility_reasons	[]
predicate_version	chi-v0.1

Field	Value
guard_crossed	True
root_multiplicity	1
confidence	1.0
pre_state_hash	ad16fb2aabdcda402c1436b1d78887b84032175f1a00431a52efd6d6d693fb97
post_state_hash	0bcb26d426ef00ebde0440b812b8a8a17a6fae8214c1887a588402a9acfaabb0
lineage_parent	0c34c34eab1a93ccece2456125c824a2
branch_created	root/route-A-to-B@2
parity_before	0
parity_after	1
invariant_report	{"passed": true, "checked": ["mass-class-A"]}

Why this example matters

- The event is solved as a root, not discovered by frame stepping.
- The event exists only because support and compatibility admit it.
- The parity bit changes but does not replace the rest of state.
- The transition has explicit before/after hashes and invariant report.
- Identity persists through lineage even though sheet, branch, parity, and address update.

18. Experiment program and benchmarks

The decisive criterion is not whether the system sounds non-raster. Compatibility plus event solving must grow more slowly than the structure the system avoids materializing. Every benchmark therefore needs an electronic or conventional reference, declared error boundaries, and a kill criterion. [S1 p. 8; S6 pp. 6, 9-10]

Software experiment matrix

Experiment	What it tests	Success evidence	Failure/kill evidence
Horizon skipping	Whether <code>state_at(t)</code> cost depends on skipped steps or expression size.	Cost remains tied to expression complexity across larger time horizons.	Expression or branch growth tracks elapsed horizon.
Compatible root solving	Whether support and chi prune candidate events enough.	Solver candidate count stays sparse; certified ordering is stable.	Event explosion or broad scan dominates.
Double-vacuum co-location	Whether same-coordinate states remain isolated until admitted.	No cross-sheet event before explicit compatibility.	Implicit coupling leaks across sectors.
Transition routing	Whether parity/sheet changes preserve invariants and lineage.	Deterministic post-state and reconstructable audit.	Ambiguous ordering, lost lineage, or invariant failure.
Grammar depth stress	Whether rule composition stays closed or normalizable.	Expression growth is bounded within declared family.	Closure fails or expressions grow without practical bound.
Split/merge identity	Whether lineage reconstructs nontrivial persistence.	All parents/children and reconciliation decisions are recoverable.	Address collisions or irreversible ancestry loss.
Projection independence	Whether multiple outputs derive from one authoritative query state.	Raster/vector/print outputs disagree only within declared projection error.	Output pipeline silently becomes the authority.

Core software metrics

Metric	Definition	Purpose
Query latency distribution	median, p95, p99 by query class	Separates typical from degeneracy/tail behavior.
Candidate reduction	relations before support / after support / after compatibility	Measures actual pruning value.
Root certification rate	certified events / attempted event queries	Tracks solver reliability.
Miss / false event rate	against exact or high-precision reference	Defines correctness boundary.
Expression growth	nodes/bytes versus grammar depth and branch history	Tests closure and normalization.
Novelty memory	stored log + branch bytes versus materialized baseline	Tests memory hypothesis.
Lineage reconstruction	success, ambiguity, time, collision count	Tests identity semantics.
Projection divergence	difference between views of same query result	Tests downstream-view discipline.

19. Optional physical bridge: Hollowlens-0

The physical path is optional and should not be confused with the software ontology. S6 translates the corpus into a conventional optofluidic experiment: a local spherical input field, a tunable liquid coupler, a 2x2 interferometric waveguide, balanced detectors, and a digital sidecar that applies support, compatibility, guard, confidence, parity, and lineage logic. [S6 pp. 1-9]

Optional physical bridge: Hollowlens-0

Bounded optofluidic implementation of the event calculus

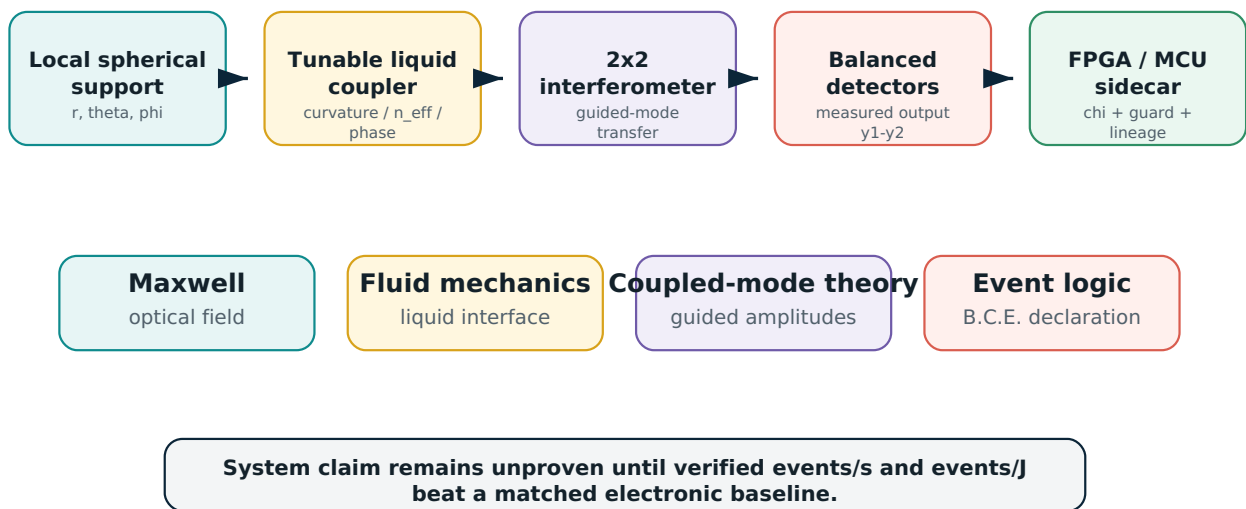


Figure 12. Hollowlens-0 is a bounded bridge from local support to a measured compatibility event.

S6 pp. 3-10.

Type-separated stack

- **Maxwell layer:** optical field in the actual dielectric geometry.
- **Liquid/interface layer:** meniscus curvature, refractive index, actuation, settling, hysteresis.
- **Guided-mode layer:** attenuation, propagation constants, coupling, phase, interference.
- **Event layer:** support, compatibility, measured guard, threshold, uncertainty, lineage.
- **Control sidecar:** calibration, drift detection, policy, timing, fallback reference.

Spherical throughput

VERIFIED EVENTS PER SECOND

$$\text{Theta_sph} = \text{N_verified} / \text{Delta_t}$$

A verified event counts only if it was inside the declared support, compatible with the selected channel, crossed the measured guard, and met the confidence threshold. Photon flux, baud rate, or propagation speed alone is not spherical throughput. [S6 p. 6]

Minimum comparison

The system-level claim remains unproven until the complete source, liquid actuator, waveguide, detector, calibration loop, and B.C.E. sidecar deliver more verified events per second and per joule than a matched electronic baseline at the same error and uncertainty boundary. [S6 pp. 9, 12]

20. Risks, failure modes, and kill criteria

The mature corpus is strongest where it admits failure. Long horizons, logarithmic coordinates, phase schedules, singular centers, near-singular operators, tangencies, coincident surfaces, rank changes, branch explosion, event explosion, and semantic grounding are not implementation footnotes. They determine whether the ontology can be executed at all. [S1 pp. 8, 11]

Risk	Failure symptom	Required response	Kill criterion
Algebraic closure	Composition leaves the solvable/normalized relation family.	Restrict grammar, normalize, or compile to certified approximation.	Expression growth eliminates direct-query advantage.
Event explosion	Too many roots or candidate relations.	Improve support/compatibility, bound interval, simplify relation family.	Solver work approaches/exceeds materialization baseline.
Degeneracy	Tangency, coincident surfaces, multiple roots, unstable ordering.	Interval methods, multiplicity tracking, tie policy, uncertainty.	Missed or unordered events exceed tolerance.
Numerical conditioning	Long-horizon phase drift, log singularity, rank instability.	Scaled coordinates, certified numerics, error bounds, reference solver.	Answers cannot be certified at required tolerance.
Branch explosion	Counterfactual history grows combinatorially.	Prune, merge with explicit policy, summarize, checkpoint.	Memory or query cost scales with all unbounded branches.
Lineage ambiguity	Splits/merges cannot be reconstructed.	Explicit parent sets and reconciliation rules.	Identity collisions or unrecoverable ancestry.
Semantic underspecification	Compatibility says "yes/no" without explainable policy.	Reason codes, versioned predicates, typed permissions.	System cannot audit why an event was admitted.
Projection feedback	Display artifacts alter authoritative state.	One-way view contract; explicit sensor-to-log events only.	Framebuffer becomes implicit state store.
Hardware drift	Optical transfer changes with temperature/aging/liquid state.	Reference channels, calibration, bounded uncertainty.	Calibration energy/time erases benefit.

Claim discipline used in the new report

Keep useful operators; separate source definition from validation.

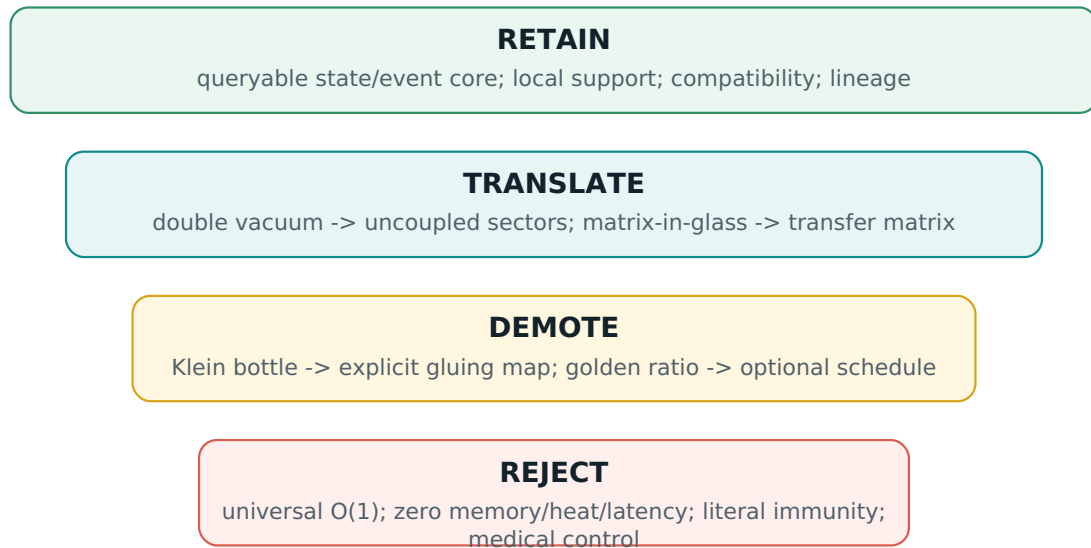


Figure 13. Claim discipline is an engineering feature, not a retreat.

The bounded sources explicitly separate component feasibility from system validation.

21. Claim-by-claim disposition

The matrix below preserves the source claims while preventing the report from presenting them at the wrong evidence level. "Retain" means the operator belongs in the architecture. "Test" means the claim becomes an experiment. "Demote" means the motif remains useful but not literal. "Reject" means it is not accepted as a requirement or established result.

ID	Claim	Assessment	Notes and source
C01	The authoritative object is a directly queryable relation/state/event substrate rather than an image or frame buffer.	canonical retain	Core interpretive rule. [S1 pp.1,5-10; S5 pp.1,9]
C02	Local spherical or radial-angular supports organize relevance, coupling, sensing, and measurement.	canonical retain	Local chart only; not global remeshing. [S1 pp.1,5-7; S6 pp.3,6]
C03	A finite grammar plus relations, transitions, invariants, and an exogenous log can define a restricted equation world.	canonical architecture retain	Prototype in bounded relation families. [S1 pp.5-8,11-12; S5 pp.10-11]
C04	SDF=0 or B=0 is best read as an event/transition surface, not a place to keep marching through.	canonical correction retain	Prevents hidden rasterization from returning. [S1 p.7; S2 pp.16-19; S5 p.54]
C05	Events require relation satisfaction, support admission, and compatibility.	canonical retain	Geometry alone is insufficient. [S1 pp.6-8; S6 pp.3,5,7]
C06	Double vacuum means co-location without coupling across incompatible phase/sheet/address sectors.	canonical retain	Absence of coupling, not literal emptiness. [S1 p.6; S5 pp.32-33; S6 pp.2,4]
C07	One bit is a routing/parity/validity flag, not the complete state.	canonical correction retain	Schema is mandatory. [S1 pp.5-7; S6 pp.2,7]
C08	Identity should persist by lineage and invariants rather than current coordinate alone.	canonical retain	Split/merge rules still required. [S1 pp.1,5-8,11; S5 pp.6-8]
C09	Closed dynamics can be regenerated from seed and grammar, while irreducible novelty is logged.	architecture hypothesis prototype	Measure memory and recomputation cost. [S1 pp.8,11-12; S5 pp.10-11]
C10	For a fixed closed expression, state_at(t) cost may be independent of skipped timesteps.	bounded complexity claim test	Not universal O(1). [S1 p.8]
C11	next_event can replace frame replay when compatible event roots are tractable.	bounded complexity claim test	Can fail from event density, degeneracy, branches. [S1 pp.7-8,11-12]
C12	Compatibility plus event solving must cost less than the materialization avoided.	canonical success criterion retain	Primary falsification rule. [S1 p.8; S6 p.6]

ID	Claim	Assessment	Notes and source
C13	Rasterization, raymarching, and ray tracing may remain downstream projections.	canonical retain	Do not mistake projector for state space. [S1 pp.9-10; S5 p.9]
C14	The KC Vector system can be interpreted as a downstream projection adapter over log-polar addresses and implicit boundaries.	projection proposal retain with caveat	Not the core substrate. [S7 pp.1-10]
C15	KC Feynman vectors are a fully specified quantum path-integral method.	unsupported terminology rename/demote	Source supplies no derivation; normalize to phase-amplitude accumulator. [S7 pp.1-3]
C16	A log-polar chart turns radial scaling into additive translation in log radius.	source-defined mathematical observation retain	Chart has singularity/conditioning issues. [S7 p.7; S3 pp.2-3]
C17	Sieve and jitter motifs describe sensor sampling, dithering, and candidate reduction.	cross-domain analogy retain as layer-specific	Not proof that all sampling can vanish. [S3 pp.13-15,24-33; S7 pp.8-10]
C18	All spatial sieves, BVHs, octrees, and broad phases are eliminated universally.	inflated claim demote/test	Only possible for restricted analytic supports with effective compatibility. [S2 pp.30-32; S5 pp.14-16]
C19	Klein bottle topology is a useful routing/gluing abstraction.	bounded abstraction retain	Must specify ports, orientation map, transfer matrix. [S1 p.6; S6 pp.2,5]
C20	A literal Klein-bottle device automatically prevents tunneling or physical interaction.	unsupported physical claim reject	Topology is not a substitute for dynamics or fabrication. [S2 pp.16-25; S5 pp.35,64]
C21	The quad hourglass is a four-sector routing visualization around a rank-changing event locus.	normalized interpretation retain	Not a topology proof. [S1 p.6; S2 pp.23-25; S5 pp.33,54]
C22	$\lambda=0$ may mark rank loss/null-space entry at an event.	normalized mathematical motif retain with tolerance	Numerical thresholds and operator definition required. [S2 pp.22-25; S5 pp.33,54]
C23	The golden ratio is required for correctness or loop avoidance.	speculative motif reject as requirement	May be an optional schedule; no proof supplied. [S2 pp.12-19; S4 pp.10-11]
C24	Phi is used consistently across the corpus.	false due to symbol overload correct	Separate angle, phase, golden ratio, and torque. [S2; S4; S5]
C25	The Ben Burgers overlap can be read as shared domain, interference, or tolerance.	geometric brainstorming retain as analogy	No unique mathematical ratio established. [S4 pp.16-19]

ID	Claim	Assessment	Notes and source
C26	Zero-based indexing changes 21+7+7 from 35 to 36.	incorrect if read as arithmetic correct	35 is an index naming the 36th position. [S4 pp.1-3]
C27	Hollowlens-0 is a bounded optofluidic demonstrator with a digital sidecar.	canonical prototype retain	Must preregister support, modes, thresholds, error and energy boundary. [S6 pp.4,7-10]
C28	Matrix-in-glass is a calibrated optical transfer function.	bounded hardware translation retain	Not a zero-cost computer. [S6 pp.2,5]
C29	Double vacuum in hardware can be implemented as uncoupled or isolated modes/channels.	bounded hardware translation retain	Measure crosstalk and false events. [S6 pp.2,4]
C30	Spherical throughput is verified events per second with support, compatibility, guard and confidence.	canonical metric retain	Report loss, energy, miss/false rates and calibration. [S6 pp.3,6,11]
C31	The integrated optical system has proven an advantage over matched electronics.	not established test	Component feasibility is not system validation. [S6 pp.9-12]
C32	The architecture has zero latency, zero heat, zero power, or zero memory.	explicitly rejected totalizing claim reject	All physical and computational layers carry cost. [S2 pp.15,26-35; S5 multiple; S6 pp.2,10]
C33	The architecture universally solves chaos or all PDEs exactly.	explicitly rejected totalizing claim reject	Restricted relation families only. [S2; S5; S6 p.10]
C34	The architecture directly controls biology, gene expression, CRISPR, protein folding, or medical outcomes.	unsupported application claim exclude	At most a simulation/design prompt; not a mechanism. [S5 pp.3-6; S6 p.10]
C35	Political, psychological, and personal narratives in later source pages are technical proof.	metaphorical narrative exclude from core	Preserve as provenance, not system mechanism. [S3 later pages; S5 pp.13-35]
C36	A person or identity can become physically untouchable through phase sheets or UUIDs.	unsupported literal claim reject	Reinterpret only as access-control/provenance metaphor. [S5 pp.14-18,32-35]
C37	The headless query-first prototype is the truest first experiment.	canonical recommendation retain	Avoid renderer-first benchmarking. [S1 pp.9,11-12; S6 p.9]
C38	A hybrid engine remains a practical bridge.	supported engineering bridge retain	Stable core plus optional rendering/volumetric stages. [S1 p.9]
C39	Numerical conditioning is part of the ontology, not an implementation afterthought.	canonical risk retain	Log coordinates, tangencies, coincident surfaces, rank changes. [S1 pp.8-9]

ID	Claim	Assessment	Notes and source
C40	The first useful benchmark is six queries: state, next event, events in cone, phase coupling, transition routing, identity rebuild.	canonical experiment plan retain	Forms the Equation World Zero exit criteria. [S1 pp.11-12]

22. Research program and recommended next build

The source corpus closes with the right instruction: formalize Equation World Zero, and begin with state, compatibility, event, transition, identity, and memory. The proof target is narrow: one nontrivial relation family that answers state-at-time and next-event without replaying every frame, under explicit compatibility, with stable lineage, bounded expression growth, and memory proportional to novelty rather than geometric extent. [S1 p. 12]

Recommended build order

1. **Freeze schemas.** Adopt the included state and event JSON schemas; version every compatibility predicate.
2. **Choose one relation family.** Begin with line/circle/quadratic guards or another family with certified roots.
3. **Implement six queries.** Keep the core headless and deterministic.
4. **Instrument every cost.** Candidate counts, root work, branch growth, memory, error, and event ordering.
5. **Add one projection adapter.** Use KC Vector as a downstream view, never as the world store.
6. **Stress topology semantics.** Test co-location, parity, sheet gluing, splits, merges, and invariant preservation.
7. **Compare against a baseline.** Frame stepping, spatial index, or event queue at equal correctness.
8. **Only then consider Hollowlens-0.** Pre-register support, mode, guard, error, energy, and calibration boundaries.

EXIT CRITERION

Proceed only if compatibility plus event solving and lineage maintenance cost less than the state/materialization work avoided, at the same correctness and uncertainty boundary. Otherwise retain the useful operators inside a conventional hybrid engine.

Package deliverables supporting the program

- Reference Python prototype and tests.
- State/event JSON schemas and YAML query API.
- Operator lexicon and normalization rules.
- Claim matrix with evidence status and disposition.
- JSON-LD and GraphML ontology.
- Source-page maps and full text extracts.
- Original seven PDFs for self-contained provenance.
- Illustrated diagrams separating core, projections, and hardware.

Appendix A - operator lexicon

This lexicon is the detailed decoder for the corpus. A status of "canonical" refers to the source-derived architecture; "normalized" refers to a disciplined translation; "projection" identifies downstream output behavior; "rejected" or "excluded" marks a boundary.

Claim Boundary

Term	Normalized meaning	Formal role	Status	Source
Universal O(1)	The mature sources reject a universal constant-cost universe. Only fixed closed expressions may have horizon-independent state evaluation.	not accepted	rejected totalizing claim	[S1 pp.1,8; S6 p.10]
Zero memory / heat / latency	Rejected as system claims. Storage, branch history, optical sources, actuation, detection, calibration, and control all carry costs.	not accepted	rejected totalizing claim	[S1 pp.1,8; S6 pp.2,10]
Biological or medical control	The corpus contains analogies and application prompts, but the bounded engineering report explicitly makes no general biological or medical mechanism claim.	not established	excluded from technical core	[S5 pp.3-6; S6 p.10]

Computer Graphics

Term	Normalized meaning	Formal role	Status	Source
Rasterization	A downstream projection that maps stored or computed geometry into pixels. The corpus rejects using the raster as the authoritative world model.	materialized image view	downstream projection	[S1 pp.1,9-10; S5 pp.1,9; S7]
Raymarching	Repeated field sampling along projected directions. It may visualize an SDF but is explicitly not the mature core.	sample f along ray	downstream / not core	[S1 pp.1,7,9-10; S2 pp.12-17]
Ray tracing	Transport or visibility path computation. It can consume the substrate but does not define the substrate.	path/visibility solver	downstream projection	[S1 pp.9-10]
Hybrid engine path	A bridge architecture with stable geometric/graph world state plus optional cone integration, volumetrics, raymarching, or display stages.	core + projection stage	supported bridge	[S1 p.9]

Computer Science

Term	Normalized meaning	Formal role	Status	Source
Headless query-first path	The preferred experiment: algebra, compatibility, events, transitions, lineage, and logs before images. Success is not visual fidelity.	query core	canonical prototype path	[S1 pp.9,11-12; S6 p.9]

Core Ontology

Term	Normalized meaning	Formal role	Status	Source
State manifold	The product space that carries position, time, phase, sheet, lineage address, and branch. It is the authoritative state space, not an image buffer.	$Q = P^n \times R_{\text{time}} \times S^1_{\text{phase}} \times Z^2_{\text{sheet}} \times A_{\text{address}} \times B_{\text{branch}}$	canonical	[S1 pp.5-6; S5 pp.1,9-11]
Finite grammar	A bounded generative rule set that produces closed dynamics and relation expressions instead of storing a complete object inventory.	G	canonical	[S1 pp.5-7,11-12; S5 pp.10-11]
Relation algebra	Declarative equations or predicates that define possible states and transition surfaces.	$R_j(q; \theta_j) = 0$	canonical	[S1 pp.1,5-7; S5 pp.1,9]
Event	A guard crossing that is admitted by support and compatibility. An event is not merely a geometric hit.	$t^* = \min\{t \geq t_0 : R_j(q(t)) = 0 \text{ and } C \leq 0 \text{ and } \chi = 1\}$	canonical	[S1 pp.6-8; S6 pp.3,5,7]
Transition	The state update applied at an admitted event while preserving declared invariants and lineage.	$q(t^+) = T_j(q(t^*), \text{context})$	canonical	[S1 pp.6-7]
Invariant	A property that must persist across transitions, coordinate changes, or reconstruction.	I	canonical	[S1 pp.1,5-7,11; S5 pp.7-11]
Compatibility predicate	A Boolean or typed predicate controlling whether co-located states, channels, or relations may couple.	$\chi(e, j, t) \in \{0, 1\}$	canonical	[S1 pp.6-8; S5 pp.13-16,32-33; S6 pp.3,5,7]

Display Hardware

Term	Normalized meaning	Formal role	Status	Source
Pulse-density modulation	A temporal 1-bit stream whose pulse density represents intensity. It is a possible output adapter for a physical display, not the state substrate.	PDM	projection-layer proposal	[S7 pp.4-6]

Event/Hardware

Term	Normalized meaning	Formal role	Status	Source
B.C.E.	Bounded Compatibility Event: support admitted, channel compatible, guard crossed, and confidence threshold met.	support and χ and $g=0$	canonical bounded operator	[S6 pp.1,3,5,7]

Geometry

Term	Normalized meaning	Formal role	Status	Source
Signed distance field	An implicit scalar field that may define geometry or guards. In the mature reading it is evaluated as a relation; raymarching is optional downstream behavior.	$f(q)$	canonical with projection variant	[S1 pp.7,9-10; S7 pp.1-2]
Torque vector	The corrected interpretation in the Ben Burgers document for twisting or folding an oval/phi-like cross-section. Use tau, not phi, to avoid symbol collision.	τ	geometric brainstorming	[S4 pp.11-14]

Geometry/Graphics

Term	Normalized meaning	Formal role	Status	Source
Log-polar chart	A local coordinate transform $\rho = \log r$, $\theta = \text{atan2}(y, x)$. It supports radial multiscale addressing but introduces singularity and conditioning risks.	$\rho = \ln r$; $\theta = \text{atan2}(y, x)$	projection /local chart	[S2 pp.12-19; S3 pp.2-3; S7 pp.1,7]

Graphics Projection

Term	Normalized meaning	Formal role	Status	Source
Phase-amplitude edge accumulator	A source-faithful normalization of the KC Feynman-vector idea: accumulate weighted phase samples near an implicit boundary, then quantize for output.	$A(p)$	decoder inference	[S7 pp.1-3]

Graphics Terminology

Term	Normalized meaning	Formal role	Status	Source
Feynman vector	The KC document uses this name for phase-amplitude vectors integrated near SDF boundaries. No formal path-integral derivation is supplied, so this report renames it a phase-amplitude edge accumulator.	$A(p) = \sum_k a_k \exp(i \phi_k)$	speculative / renamed for clarity	[S7 pp.1-3]

Graphics/Indexing

Term	Normalized meaning	Formal role	Status	Source
1-bit log-polar LUT	In exploratory sources this alternates between active-sector mask, parity field, and output quantizer. The report separates these roles rather than treating one bit as the whole state.	$\text{mask}(\rho, \theta)$	role-overloaded proposal	[S2 pp.12-19; S7 pp.1-5]

Graphics/Output

Term	Normalized meaning	Formal role	Status	Source
Anisotropic dithering	Log-radius- and angle-aware stochastic quantization used for screen or print output.	blue-noise threshold	projection-layer proposal	[S7 pp.5,8-10]
Prepress adapter	A static 1-bit halftone path: CMYK mapping, log-polar blue noise, SDF edge lock, and dot-gain compensation.	1-bit TIFF/PDF	projection-layer proposal	[S7 pp.8-10]

Hardware

Term	Normalized meaning	Formal role	Status	Source
Matrix-in-glass	A calibrated passive or tunable optical transfer matrix in a waveguide mesh. It is not a stored picture or zero-energy computer.	optical transfer matrix	bounded hardware translation	[S5 pp.47-50; S6 pp.2,5]

Term	Normalized meaning	Formal role	Status	Source
Liquid substrate	A liquid lens, overlaid, or core coupled to a solid photonic substrate to tune focus, effective index, phase, or mode coupling.	$n_{eff}(u)$	bounded hardware translation	[S5 pp.58-64; S6 pp.2-5,8]

Hardware/Control

Term	Normalized meaning	Formal role	Status	Source
Digital sidecar	Conventional electronics for calibration, thresholds, uncertainty, lineage, policy, drift detection, and fallback. Its necessity prevents magical zero-latency claims.	FPGA/MCU	canonical control element	[S6 pp.4,7-10]

Information Science

Term	Normalized meaning	Formal role	Status	Source
Exogenous log	A record of irreducible external novelty that cannot be regenerated from seed and grammar.	L	canonical	[S1 pp.6,8,11-12; S5 pp.10-11]
Novelty-proportional memory	Memory discipline in which closed dynamics are regenerated and storage grows with external novelty and necessary branch history rather than geometric extent alone.	$M \sim \text{seed} + \text{grammar} + \text{log} + \text{branch history}$	bounded hypothesis	[S1 pp.8,11-12; S5 pp.10-11]
Lineage	Persistent provenance through splits, merges, transitions, and coordinate changes.	$\text{lineage}(e)$	canonical	[S1 pp.1,5-8,11; S5 pp.6-8,11]
Ontological address	A generative identity derived from seed, grammar path, and invariant history; normalized here as a provenance key rather than a literal magic UUID.	$a_e = H(\text{seed}, \text{path}, \text{lineage})$	normalized operator	[S1 pp.5-6; S5 pp.6-8]
Branch	A version, fork, or counterfactual path in the state evolution. Branch history is a real cost and can explode.	b_e	canonical	[S1 pp.5-8,11]
Zero-based timeline	A reference-frame or indexing shift: an offset from index 0 identifies positions differently from 1-based counting. It does not change arithmetic.	index 0	corrected computing motif	[S4 pp.1-6]
Materialized view	A rendered image, mesh, cache, or sampled field derived from the authoritative relation/event state. This term is the report's information-science translation.	$\text{view}(Q, \text{query})$	decoder inference	[S1 pp.1,9-10; S5 p.9]

Information/Graphics

Term	Normalized meaning	Formal role	Status	Source
Sieve	A filter or candidate-reduction stage: aperture, receptor mosaic, BVH, octree, mask, or compatibility filter. The corpus aims to replace broad materialization with analytic support and compatibility, not prove all indexing unnecessary.	filter/index	normalized caution	[S3 pp.13-15,24-33; S5 pp.14-16; S2 pp.30-32]

Linear Algebra

Term	Normalized meaning	Formal role	Status	Source
Eigenvector	An invariant direction under a transformation, used as an anchor or stable mode.	$A v = \lambda v$	normalized operator	[S2 pp.20-25; S5 pp.2,17,51]
Null space	The inactive or rejected subspace produced by a compatibility/kernel operation. It should mean algebraic exclusion, not physical disappearance.	$\ker A$	normalized operator	[S2 pp.22-25; S5 pp.15,33]

Linear Algebra/Event

Term	Normalized meaning	Formal role	Status	Source
Eigenvalue zero	A rank-loss or null-space event used metaphorically as an off-state. In an implementation it must be tied to a defined operator and numerical tolerance.	$\lambda \approx 0$	normalized with numerical guard	[S2 pp.22-25; S5 pp.33,54]

Method

Term	Normalized meaning	Formal role	Status	Source
Type safety	Keep electromagnetics, fluid mechanics, coupled-mode theory, event logic, topology, and information semantics in their own domains with explicit interfaces.	domain separation	canonical engineering discipline	[S6 pp.2,5]

Metrics

Term	Normalized meaning	Formal role	Status	Source
Spherical throughput	Verified compatible events or answered queries per second at a declared error budget, not raw photon flux or propagation speed.	$\Theta_{\text{sph}} = N_{\text{verified}} / \Delta t$	canonical metric	[S6 pp.3,6,11]

Notation

Term	Normalized meaning	Formal role	Status	Source
Phi: angular coordinate	The local angular coordinate or lighthouse incidence angle. Distinguish from phase schedule and golden ratio.	ϕ_{angle}	ambiguous source symbol	[S2 pp.12-19; S5 pp.2-3]
Phi: phase law	A phase coordinate or schedule used in compatibility and coupling. Irrational schedules are presented as heuristics, not proven requirements.	$\phi_{\text{phase}}(t)$	canonical concept, heuristic schedule	[S1 pp.5-6; S2 pp.14-19; S5 pp.2-3]

Notation/Metaphor

Term	Normalized meaning	Formal role	Status	Source
Golden ratio	An optional scaling or phase motif in exploratory sources. The mature architecture does not require the golden ratio.	$\phi_g = (1 + \sqrt{5})/2$	speculative motif	[S2 pp.12-19; S4 pp.10-11]

Prototype

Term	Normalized meaning	Formal role	Status	Source
Hollowlens-0	A minimum optofluidic demonstrator: scanned input, tunable liquid coupler, 2x2 interferometer, balanced detectors, digital sidecar, and pre-registered event tests.	prototype stack	canonical prototype	[S6 pp.4,7-10]
Equation World Zero	Restricted headless prototype with low-dimensional homogeneous coordinates, continuous time, two sheets, one orientation bit, bounded relation families, finite grammar depth, lineage, and symbolic outputs.	Ew_0	canonical prototype	[S1 pp.7,11-12]

Query Api

Term	Normalized meaning	Formal role	Status	Source
State-at-time query	Direct evaluation of a closed state expression at time t without replaying every intermediate frame. Horizon independence is only plausible for fixed bounded expressions.	$state_at(e, t)$	canonical bounded claim	[S1 pp.7-8,10-12]
Next-event query	Find the earliest compatible relation crossing after t0. Cost is compatibility plus root/event solving plus branch resolution.	$next_event(e, t_0)$	canonical	[S1 pp.7-8,10-12]
Events-in-support query	Enumerate admitted events inside a local radial-angular or causal support.	$events_in_support(C_alpha, t_0, t_1)$	canonical	[S1 pp.6-7,11]

Routing

Term	Normalized meaning	Formal role	Status	Source
One-bit parity	A narrow route, orientation, validity, freshness, or availability flag. It is not the full world state.	$p \leftarrow NOT\ p$	canonical narrow role	[S1 pp.5-7; S5 pp.33,44-47,54; S6 pp.2,7]

Routing/Topology

Term	Normalized meaning	Formal role	Status	Source
Pinch point	A rank-changing event locus at the center of the hourglass routing visualization. It becomes meaningful only with an explicit guard and transition map.	$B=0$	normalized operator	[S1 p.6; S2 pp.24-25; S5 pp.33,54]
Quad hourglass	A four-chamber routing partition with a central event locus and sheet/orientation transitions. It is a disciplined visualization, not a topology proof.	four chambers + pinch	normalized operator	[S1 p.6; S2 pp.23-25; S5 pp.32-33,54]

Sensing/Graphics

Term	Normalized meaning	Formal role	Status	Source
Spherical LUT	A radial-angular lookup or local sensory organization. It should be read as a local chart/index, not proof that atoms or the whole world are spherical records.	$LUT(r, \theta, \phi)$	metaphoric al/local support	[S3 pp.14-20,24-33]
Jitter / micro-saccade	Controlled perturbation used to fill sampling gaps, dither quantization, or estimate sub-cell structure. It belongs to sensors and projections, not the invariant core.	$\epsilon(t)$	projection/sensing operator	[S3 pp.13-15,24-33; S7 pp.8-10]

Topology

Term	Normalized meaning	Formal role	Status	Source
Klein bottle	An orientation-reversing gluing or routing abstraction. A usable implementation must specify ports, transition functions, orientation map, and transfer matrix.	quotient/ gluing map	demoted from literal physics	[S1 pp.2,6; S2 pp.7-10,16-19; S6 pp.2,5,10]

Topology/Event

Term	Normalized meaning	Formal role	Status	Source
SDF zero / B=0	The explicit transition or guard surface where event semantics are invoked. It should not be confused with continuing to sample through an SDF.	$B(q)=0$ or $f(q)=0$	canonical	[S1 pp.5-7; S2 pp.16-19; S5 p.54; S6 pp.3,5,7]

Topology/Information

Term	Normalized meaning	Formal role	Status	Source
Double vacuum	Phase-separated co-location: two states may share spatial coordinates yet fail to interact because phase, sheet, orientation, address, or policy are incompatible. The second vacuum is absence of coupling.	same x ; $\chi=0$	canonical	[S1 p.6; S2 pp.14-19; S5 pp.32-33; S6 pp.2,4]
Phase sheet	A sector or namespace that labels co-located states and participates in compatibility.	σ in Z^2 or finite sheet set	canonical	[S1 pp.5-6; S5 pp.13-15,32-33]
Overlap / weld zone	The slight overlap of folded D/ϕ shapes is read as intersection, tolerance buffer, constructive interference, or shared domain $A \cap B$.	$A \cap B$	metaphorical geometric operator	[S4 pp.16-19]

Topology/Metaphor

Term	Normalized meaning	Formal role	Status	Source
Mobius strip	A one-sided continuity and inversion motif used in the optical/art narrative. It can motivate orientation reversal but is not required by the formal core.	local orientation flip	metaphorical / optional	[S3 pp.5-10]

Topology/Query

Term	Normalized meaning	Formal role	Status	Source
Local spherical support	A local chart for radial reach, angular support, orientation, field of view, uncertainty, or domain of dependence. It is not a global spherical remeshing.	$C_{\alpha}(q, t) \leq 0$	canonical	[S1 pp.1,5-7,9-10; S6 pp.3,6]
Cone	An analytic relevance, causal, sensing, collision, memory, or grammar support. It is not intrinsically a ray bundle.	C_{α}	canonical normalized operator	[S1 pp.1,5-7,10; S2 pp.14-19]
Shell	A radial support boundary or nested relevance band around a local origin.	$r \in [r_0, r_1]$	normalized operator	[S1 pp.1,5-6]

Appendix B - source page map

The per-document maps in the ZIP provide longer notes. This condensed table identifies the page clusters that control the report.

Source	Pages	Topic	Technical reading
S1	p.1	Interpretive shift	Projection tools are downstream; ontology is queryable relations/events/phases/identities.
S1	pp.5-6	Formal core	Product state space, event equation, double vacuum, routing visualization.
S1	pp.7-8	Prototype and complexity	Equation World Zero, bounded horizon independence, numerical risks.
S1	pp.9-10	Graphics boundary	Rasterization/raymarching/tracing contrasted with query semantics.
S1	pp.11-12	Research program	Bottlenecks, six required queries, exit criteria.
S2	pp.14-17	User correction and analytic core	No raster/marching; analytic solids, SDF roots, phase and parity.
S2	pp.18-25	Event/routing motifs	Lighthouse around zero, phase layers, eigenspace, quad hourglass.
S2	pp.26-33	Grammar/memory proposals	Kinematic calculus, procedural grammar, CCD, UUID and compact-state hypotheses.
S3	pp.1-3	Optical encoding	Spectra, polar harmonics, logarithmic compression, stacked LUTs.
S3	pp.11-15	Sieve and jitter	Device-mediated scale, receptor mosaics, micro-jitter and sampling gaps.
S3	pp.16-20	Spherical field proposal	SDF beyond zero, spherical LUT, grammar, frontend CCD motivation.
S3	pp.30-33	Self-correction	A hard event surface is needed or hidden sampling returns.
S4	pp.1-3	Zero indexing	Offset 35 from zero is the 36th position; arithmetic remains 35.
S4	pp.16-19	Fold/overlap	Two D/phi shapes, hinge, overlap as intersection, tolerance, or interference.
S5	pp.1-2	Cross-domain synthesis	Relation algebra, state/event calculus, phase and local spherical support.
S5	pp.6-11	Identity and memory	Lineage address, Person(t) != Snapshot(t), grammar, log, branch history.
S5	pp.32-33	Double vacuum	Same position/different sector, compatibility gate, parity and pinch routing.
S5	pp.47-50,58-64	Hardware prompts	Etched glass, photonics, liquid lensing, matrix-in-glass.
S6	pp.1-2	Engineering normalization	Retain operators; demote literal topology; reject zero-cost claims.
S6	pp.3-7	Bounded architecture	Signal path, governing physics, throughput, B.C.E. control.
S6	pp.8-10	Fabrication and kill criteria	Conventional process, Hollowlens-0, risk register and non-claims.
S7	pp.1-3	Projection adapter	Log-polar address, SDF/CSG, phase-amplitude edge accumulation, 1-bit output.
S7	pp.4-10	Display and print	PDM, anisotropic dither, chromatic offsets, CMYK/prepress and dot gain.

Source key

ID	Filename	SHA-256
S1	Chronological Synthesis of the Spherical Substrate Line(3).pdf	9bacfe071a64951fb412798a036e08f9d445b83087acb001b660e64d6036e052
S2	hollowland-double-vacuum(7).pdf	69a6b341d02c87f43dcaf048489ef0b046c9a6d0a24efbc61f180f696d9912a8
S3	dmt-spirit-molecule-(leeuwenhoek-animalcules-peter-the-great-netherlands-signify-phillips-netherlands) (1).pdf	43e3accd3596e6d822a5daa73e9781f05d6d47b5136071b991a977a135bdbaf c
S4	21BenBurgersStrikesBackTelNetNiet.pdf	1ab4f04669ded1aa2308c28ce634e487fa4c26ba0a82bfe436e606388ef0f6dd
S5	ADDominUS(1).pdf	2a9b755257c8c049463aefae3762139fdb2833c837f037219b236abb061c06d
S6	Spherical_Throughput_Practical_Waveguide_Liquid_Substrate_Lensing(sqwit (super quick wit).pdf	949d7ee3cd7c9e4ffdb27d63b2805ce4f1491568a6fdd5b7e04331d3c96d16ac
S7	KC Vector Graphical Art System.pdf	6518f7f73f46ec6c6288fe93f95e8d9e4f331376aee33ceb26f4508a1e30e58c

Appendix C - source-page evidence snapshots

These snapshots show the actual source pages behind the mature interpretation, bounded hardware translation, and downstream graphics adapter. They are included for visual traceability; the full PDFs and page-separated text are in the ZIP.

 ChatGPT

Chronological Synthesis of the Spherical Substrate Line

Executive Summary

The newest PDF does not merely update the archive; it changes the interpretive rule for the whole corpus. Its core instruction is to stop judging the archive as if it were fundamentally a proposal about rasterization, ray marching, voxel storage, or a conventional frame loop. Instead, it argues that those are downstream projection tools, while the actual proposal is an ontology of directly queryable relations, events, phases, and identities. In that reading, the archive's "cone," "SDF + 0," "one-bit," "double vacuum," and "hourglass" motifs are not primarily graphics primitives. They are operators in a shared state space. That is the decisive shift in the corpus. [filecite:turnOffile0L20-L33](#)

Read chronologically, the archive moves through a consistent arc. It begins with reaction-diffusion balance, collapse, gradients, and distributed state; then introduces cone and polar language; then develops a layered shared ontology for geometry, sensing, uncertainty, and semantics; then returns obsessively to boundary, parity, persistence, and continuity-through-zero; and finally, in the 19 July documents, recasts the whole project as a non-raster equation-world built from compatibility, event solving, transition rules, lineage, and an irreducible event log. The latest PDF itself summarizes that progression explicitly and treats its own contribution as a constructive correction: formalize the concept on its own terms before comparing it to established engines or renderers. [filecite:turnOffile0L54-L76](#) [filecite:turnOffile0L593-L604](#)

The strongest unified concept is therefore **not** "replace all world structure with spheres" and **not** "render the world by spherical raymarching." It is a more specific claim: a finite grammar can generate a continuous, phase-layered, event-driven state manifold in which local spherical or radial-angular supports organize relevance, coupling, and measurement, while identities are preserved by lineage and invariants rather than by instantaneous coordinates. In that concept, "spherical" is best understood as the local substrate of support, shelling, orientation, and radial reach, not as a universal replacement for every representation in the system. This interpretation matches the archive's own mid-July corrections, which repeatedly reject both "rasterized-everything" and "polar-everything" in favor of a layered or query-driven ontology. [filecite:turnOffile0L95-L118](#) [filecite:turnOffile0L258-L301](#) (Polar Coordinates as the Missing Local Chart for the Aerosol Cone Architecture.pdf, pp. 1-3) (Unified Architecture for Aerosol Cone Sensing, SDFs, Nested Shells, and Spherical Kinematics.pdf, pp. 1-2) (Hybrid Ontology for Unity 6.3.pdf, pp. 1-2)

On feasibility, the corpus is internally strongest when narrowed. The latest PDF is explicit that the most defensible form is a restricted substrate that can answer state-at-time and next-event queries without universal frame stepping, but only for controlled relation families. It also explicitly demotes the grander claims—universal O(1), zero memory, world-equals-general-AI—from doctrine to research hypotheses. That is the right reading of the corpus. Under that narrowed interpretation, the "spherical substrate line" is feasible as a prototype-worthy computational medium. As a universal replacement for graphics, simulation, memory, cognition, and semantics all at once, it remains unproven. [filecite:turnOffile0L40-L49](#) [filecite:turnOffile0L366-L375](#) [filecite:turnOffile0L579-L592](#)

1

S1 p. 1 - decisive ontology/query-first interpretation.

$$q_e(t) = (x_e(t), t, \phi_e(t), \sigma_e(t), a_e, b_e)$$

$$W = (G, R, T, I)$$

$$R_j(q; \theta_j) = 0, \quad C_a(q, t) \leq 0, \quad \chi(e, j, t) \in \{0, 1\}$$

$$t' = \min\{t \geq t_0 : R_j(q_e(t)) = 0, C_a \leq 0, \chi = 1\}$$

$$q_e(t'') = T_j(q_e(t'), \text{context})$$

In plain language, a world state is not only "where in space," but also "when," "in what phase," "on which sheet," "at which lineage-address," and "on which branch." Events are not pure geometry intersections; they happen only when a relation is satisfied, the state lies inside the relevant support cone, and the co-located sectors are compatible. [filecite:turnOffile0L202-L230](#)

Architecture of the equation world from the latest PDF, page 6

The architecture diagram on page 6 makes the ontology especially clear: grammar and exogenous log feed a projective-phase manifold; relations and compatibility operate inside that manifold; cone/hourglass queries select relevance; and transitions update state while preserving invariants. The same state description can then answer state-at-time, next-event, reachable-relation, identity, and agent-action queries. That is why the through line is deeper than graphics. It proposes a common substrate for physics-like transitions, sensing, persistence, and agency. [filecite:turnOffile0L187-L197](#)

The specifically **spherical** part of the line needs to be stated carefully. The archive's strongest correction, visible already in the mid-July analyses, is that spherical or polar structure belongs primarily to **local support, egocentric sensing, shell organization, radial reach, and phase coupling**—not to a global remeshing of all world state. *Polar Coordinates as the Missing Local Chart for the Aerosol Cone Architecture and Unified Architecture for Aerosol Cone Sensing, SDFs, Nested Shells, and Spherical Kinematics* both insist on that distinction: "keep the world model hybrid," "make the sensor model polar," and treat spherical coordinates as native to the local manifold of cones, shells, and bilateral kernels rather than as a total world replacement (Polar Coordinates... pp. 1-3; Unified Architecture... pp. 1-2). *Hybrid Ontology for Unity 6.3* makes the same point more bluntly: neither raster-everything nor polar-everything is the right reading; persistent world state, local sensing, and display should remain distinct layers (Hybrid Ontology... pp. 1-2).

The archive's best-developed singular notion is **double vacuum**. In the latest PDF it is given its cleanest formulation: two states can share the same coordinate x yet still fail to interact because their phase, sheet, orientation, or address are incompatible. The "second vacuum" is therefore not a stronger emptiness; it is the absence of coupling. This is one of the core places where the corpus stops being about visual metaphors and becomes a computational ontology. Same position does not imply same sector. Interaction becomes explicitly gated. [filecite:turnOffile0L235-L252](#)

Operational reading of double vacuum and the quad hourglass from the latest PDF, page 8

That page 8 diagram is useful because it shows how several recurrent archive motifs collapse into one structure: the four chambers become a routing partition, the central pinch becomes a rank-changing event locus, the vertical line becomes an invariant mode or eigen-axis, and the swept arc becomes phase-controlled coupling. The image is not literal topology proof, but it is a disciplined visualization of the actual computational claim. [filecite:turnOffile0L238-L243](#) [filecite:turnOffile0L245-L252](#)

6

S1 p. 6 - formal equations and double-vacuum/hourglass reading.

Prototype experiment	What it tests	Evidence of success
Minimal agent integration	Whether action, memory, and perception can reuse the same relation substrate	Agent queries and action selection operate in the same transition language as embodiment

Those experiments are not speculative additions by me: they directly condense the latest PDF's own evaluation program and exit criteria. [filecite\[turnOff\]0\[.L512-L559\]](#)

A clean prototype flow looks like this:



The most important recommendation for a proof of concept is therefore to **refuse two temptations**. The first temptation is to turn the experiment immediately into a renderer benchmark. The second temptation is to preserve the archive's most inflated claims as requirements. The proper goal is narrower: show that one nontrivial relation family can answer state-at-time and next-event queries without stepping every frame, under explicit compatibility, with stable lineage, bounded expression growth, and memory proportional to novelty rather than to geometric extent. If that works, the prototype has validated the core of the spherical substrate line. If it fails, the failure will be informative in exactly the right place. [filecite\[turnOff\]0\[.L544-L559\]](#)

The corpus's own final decision is the correct closing judgment: **GO for formalization and Equation World Zero, but do not begin by recreating a conventional scene or image pipeline. Begin with the algebra of state, compatibility, event, transition, identity, and memory.** [filecite\[turnOff\]0\[.L588-L592\]](#)

S1 p. 12 - headless prototype flow and final recommendation.

ADDominUS enters the source basis

NEW SOURCE RECORD

S8 - ADDominUS

65-page exploratory dialogue and source bundle. The decisive waveguide material appears on pp. 47-48, 50, 58, and 60-64.



SOURCE / NOT VALIDATION

WHAT IT CONTRIBUTES

A hardware-facing extension

- Etched-glass and silicon-photonc waveguides as an optional physical endpoint.
- A tunable liquid-lensing region between a spherical input field and a guided-mode network.
- "Matrix-in-glass" as a calibrated optical transfer function rather than a stored picture.
- A pinch point as a measurable threshold, dark port, cutoff, or critical-coupling event.
- One-bit parity as a route or validity flag, not as the complete state.
- Manufacturing contexts: spectroscopy, vibration isolation, microfluidics, and photonic packaging.

CORPUS TRANSLATION RULE

Keep the implementable operator; demote the totalizing claim

ADDominUS motif	Practical translation	Status
Matrix-in-glass	Calibrated passive or tunable transfer matrix in a waveguide mesh.	RETAIN
Liquid spherical field	Microfluidic lens, overlaid, or liquid-core region that changes focus and effective index.	TRANSLATE
Double vacuum	Orthogonal, uncoupled, or deliberately isolated modes/channels.	RETAIN
Klein SDF waveguide	Explicit routing/gluing map in the model; literal non-orientable hardware remains frontier.	DEMOTE
Zero tunneling	Bound the missed-event probability with continuous models and measured timing.	REWRITE
Zero latency / heat / memory	Include laser, detector, actuator, control, calibration, and storage costs.	REJECT

TYPE SAFETY

Maxwell governs the optical field. Fluid mechanics governs the liquid interface. Coupled-mode theory governs guided amplitudes. Event logic governs when a result is declared.

Source basis: ADDominUS pp. 47-48, 50, 58, 60-64; Spherical Throughput pp. 2-4; The Spherical Line, decisive-line and translation sections.

Practical waveguide liquid-substrate lensing | ADDominUS corpus integration | July 2026

S6 p. 2 - retain/translate/demote/reject hardware translation table.

The equations that replace metaphor

1. OPTICAL FIELD

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) - k_0^2 \epsilon(\mathbf{r}, u) \mathbf{E} = 0$$

Solve the electromagnetic field in the actual dielectric geometry. The actuator state u changes the permittivity through the liquid shape or index.

2. GUIDED-MODE EVOLUTION

$$\frac{\partial \mathbf{a}}{\partial z} = -\left(\frac{\alpha}{2} + \beta \mathbf{a}\right) \mathbf{a} - \sum_{j \neq 1} \kappa_{mj}(u, \lambda) \mathbf{a}_j$$

Coupled-mode theory tracks attenuation, propagation constants, and tunable coupling between modes. This is the practical "matrix-in-glass".

3. LIQUID INTERFACE + PHASE

$$\Delta p = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right), \quad \Delta \phi(u, \lambda) = \frac{2\pi}{\lambda} \int \Delta n(u) d\mathbf{r}$$

Young-Laplace links pressure to meniscus curvature. The effective-index change creates the programmable optical phase.

4. TRANSIENT ACTUATION

$$\rho \partial_t \mathbf{v} + \nabla \Pi = -\nabla p + \mu \nabla^2 \mathbf{v} + \mathbf{f}_{\text{ext}}, \quad \nabla \cdot \mathbf{v} = 0$$

Use Navier-Stokes only when the lens shape is moving quickly enough that quasi-static curvature is insufficient. Light itself remains governed by Maxwell.

5. SPHERICAL-TO-MODE COUPLING

$$\eta_k = \frac{\left| \int_{\Omega} \mathbf{E}_k \cdot \mathbf{E}_0 d\Omega \right|^2}{\left(\int_{\Omega} |\mathbf{E}_k|^2 d\Omega \right) \left(\int_{\Omega} |\mathbf{E}_0|^2 d\Omega \right)}$$

The overlap integral measures how much of the admitted radial-angular field couples into the chosen guided mode.

6. BOUNDED COMPATIBILITY EVENT

$$t^* = \inf \{ t > t_0 : S_k = 1, \chi_k = 1, g_k(y, t) = 0 \}$$

A transition exists only when the input lies in support, the channel is compatible, and the measured guard reaches zero.

TOPOLOGY DISCIPLINE

A "Klein bottle" can be retained as a routing or gluing abstraction only after its ports, orientation map, and transfer matrix are specified. It is not a substitute for Maxwell, coupled-mode theory, or fluid mechanics.

Equations are the bounded engineering formalization of the corpus. They are not extracted verbatim from ADDomHUS, whose displayed formulas are incomplete in the source export.

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Hollowlens-0

MINIMUM BUILD

One angularly scanned input beam, one tunable liquid coupler, one 2x2 interferometric waveguide, two balanced detectors, and one digital sidecar. The query is: did an input from the declared spherical sector couple into the compatible mode and cross the output threshold inside the requested time window?

BENCH LAYOUT



EXPERIMENT MATRIX

Test	Controlled change	Readout	Success criterion
Support	Move input across tetragh sectors.	Coupling map and candidate count.	Declared sector is admitted; outside sectors are suppressed.
Liquid tuning	Sweep pressure/voltage/temperature.	Phase, focus, etc. c. settling time.	Repeatable transfer with bounded hysteresis.
Compatibility	Change wavelength, polarization, mode, or phase tip.	Accepted vs rejected events.	False accept and false reject meet preset limits.
Event timing	Drive output through threshold at known times.	B.C.E. timestamp and confidence.	No missed crossings within declared bandwidth.
Energy	Measure laser, actuator, detector, and controller.	Joules per verified event.	Beats the electronic reference at equal accuracy.
Drift	Cycle temperature and run duration.	Transfer error and recalibration interval.	Calibration burden does not erase the advantage.

PRE-REGISTER BEFORE POWERING THE LASER

Wavelength, optical power, spherical support, compatible modes, thresholds, allowed uncertainty, error targets, energy accounting boundary, calibration interval, and the electronic baseline must be fixed before results are inspected.

The first proof is headless and query-based: a stable, bounded relation family that answers state/event questions without a frame loop.

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Failure modes and kill criteria

RISK REGISTER

Risk	Failure symptom	Mitigation	Kill criterion
Optical loss	Signal vanishes through couplers, fluid, bends, or interfaces.	Short path, better coupling, lower-loss materials, balanced detection.	No energy or accuracy advantage at required power.
Liquid instability	Bubbles, evaporation, wetting drift, contamination, or hysteresis.	Degas, seal, select compatible fluids, monitor pressure and temperature.	Transfer cannot be reproduced or calibrated.
Slow actuation	Lens settling is slower than the event stream.	Use quasi-static use case, faster actuator, smaller chamber, predictive control.	Required event rate exceeds physical tuning bandwidth.
Mode crosstalk	Incompatible channels leak into each other.	Mode filters, polarization control, phase trim, guard bands.	False-event rate exceeds declared tolerance.
Calibration drift	Temperature or aging changes the transfer matrix.	Reference channels, periodic calibration, digital compensation.	Calibration energy/time erases the benefit.
Event ambiguity	Threshold tangency, ringing, or detector noise creates unstable events.	Hysteresis, interval estimates, confidence, debounce, fallback sampling.	Missed or unordered events at application tolerance.
Packaging failure	Leaks, delamination, swelling, or alignment loss.	Reliability cycling, compatible bonding, mechanical relief, serviceable design.	Cannot pass intended environment or cycle count.
No system advantage	The digital sidecar does nearly all useful work.	Measure full system; simplify optical kernel.	Conventional electronic reference is cheaper, faster, or more accurate.

CLAIMS THIS INTEGRATION DOES NOT MAKE

- No zero latency: propagation is fast, but coupling, liquid actuation, detection, conversion, and control take time.
- No zero heat or zero power: lasers, modulators, actuators, detectors, and electronics dissipate energy.
- No universal $O(1)$, zero-memory universe, exact chaos, or topological immunity from missed events.
- No literal double vacuum or physical Klein-bottle proof unless a fabricable geometry and transfer map are specified.
- No general AI compression, medical effect, biological mechanism, or replacement for all renderers and PDE solvers.

This bounded interpretation follows the corpus rule: retain useful operators, require a digital reference, and treat hardware claims as frontier until measured.

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S6 p. 10 - risk register and explicit non-claims.

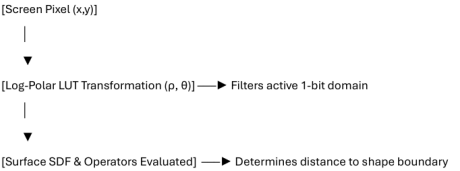
- **Geometric Operators:** To construct complex forms, basic shapes are combined using algebraic operators. For example:
 - **Union:** $\sqrt{\min(f_1(\vec{p}), f_2(\vec{p}))}$
 - **Intersection:** $\sqrt{\max(f_1(\vec{p}), f_2(\vec{p}))}$
 - **Subtraction:** $\sqrt{\max(f_1(\vec{p}), -f_2(\vec{p}))}$
- **The Zero-Isovalue Surface:** The boundary where $(f(\vec{p}) = 0)$ explicitly marks the target geometry's surface. Points where $(f(\vec{p}) < 0)$ represent the interior of the shape.

3. Pixel Evaluation Using Feynman Vectors

The unique mechanism of the Kees Klootwijk system relies on **Feynman-style path vectors** to resolve sub-pixel boundaries, aliasing, and field probabilities. Rather than evaluating the SDF as a static point, the system treats pixel filling as a quantum-amplitude superposition. [1, 2]

- **Vector Superposition:** For each active cell in the 1-bit LUT, the system assigns a state vector (a Feynman amplitude vector carrying magnitude and phase). The vector's phase is driven by the spatial gradient of the surface SDF $(\nabla f(\vec{p}))$, while its magnitude corresponds to the proximity to the $(f(\vec{p}) = 0)$ boundary. [1]
- **Path Integration for Rasterization:** The final state of a pixel is determined by summing (integrating) these vectors across sub-pixel paths. If the destructive and constructive interference of the Feynman vectors yields a net amplitude above the 1-bit threshold, the pixel is drawn as active (1) ; otherwise, it remains blank (0) .

Summary of the Rendering Pipeline



S7 p. 2 - SDF/CSG and source-defined vector accumulation language.

- **Pulse-Density Modulation (PDM):** Standard digital screens often use Pulse-Width Modulation (PWM) running at fixed frequencies (e.g., 240Hz to 4kHz) to manage brightness. The Klotwijk architecture bypasses uniform PWM intervals in favor of high-frequency PDM bitstreaming. [1]
- **The Vector Pulse:** The LEDs are driven by a continuous 1-bit stream pulsed at the maximum refresh rate of the panel's driver IC. The density of these 1-bit pulses over time directly corresponds to the magnitude of the integrated Feynman vectors. This produces smooth gradient paths and eliminates the "stair-stepping" typical of rasterized vector approximations.

3. Log-Polar Anisotropic Spatial Dithering

Because a physical LED panel features a fixed, square Cartesian lattice, mapping a logarithmic-polar spatial coordinate grid onto it creates spatial distortions (pixels near the center have high log-polar data density, while edge pixels have low density).

The LED modulation compensates for this spatial disparity through **Anisotropic Delta-Sigma Modulation**:

- **Inner Core (High-Density):** Near the log-polar origin, the data density exceeds physical pixel density. The system rapidly modulates adjacent LEDs out-of-phase with each other (spatial destructive interference), preventing visual clustering and maintaining tight vector lines.
- **Outer Edge (Low-Density):** Toward the outer edges of the screen, the log-polar sectors expand across multiple physical LEDs. The system modulates groups of neighboring LEDs with a synchronized, constructive temporal pulse stream. This smoothly diffuses the vector across the physical gaps, mimicking a continuous analog beam deflection (similar to an oscilloscope).

Systemic Data Path for LED Control

[Feynman Vector Integration Engine] —> Yields high-precision phase & amplitude

|



[Anisotropic Delta-Sigma Modulator] —> Balances log-polar density vs. Cartesian layout

|



S7 p. 5 - 1-bit display modulation and anisotropic mapping.

1. Posturization of Feynman Amplitudes

In the print-prep stage, the continuous vector interference values must be clamped into strict, binary 1-bit values (Ink or No Ink / Cyan, Magenta, Yellow, Black plates) while preserving the geometric intent of the original surface SDF.

- **Probability Thresholding:** The magnitude of the integrated Feynman vectors at any given coordinate represents a local probability density. The posturizer converts this density into a binary state (0 or 1) using a deterministic, blue-noise threshold matrix.
- **SDF Edge Preservation:** Near the shape boundaries ($\|\nabla f(\vec{p})\| \approx 0$), the posturizer locks the 1-bit output to match the precise orientation of the spatial gradient $\|\nabla f(\vec{p})\|$. This ensures that vector edges remain razor-sharp and do not suffer from the "fuzzy" or pixelated degradation common in standard raster-to-print conversions.

2. Log-Polar Blue Noise Jitter

Standard print dithering uses a square Cartesian grid. If applied to Klotwijk's log-polar art, the expanding spatial sectors toward the outer edges would cause the dot density to unnaturally thin out, ruining the tonal consistency.

- **Logarithmic Dot Scaling:** The jitter engine scales the spatial frequency of the noise dynamically based on the log-radius p . Near the origin (center), the dots are ultra-fine and densely packed. Toward the outer margins, the dot size adjusts proportionally to the log-polar coordinate expansion.
- **Anisotropic Orientation:** The jitter is stochastically distributed but structurally oriented along the angular vector θ . This creates a directional, sweep-like grain structure that visually reinforces the concentric and radial nature of the underlying math, making the print dot pattern look organic and intentional rather than like digital artifacting.

3. Prepress Plate Alignment & Ink Clumping Limits

When translating the wavelength-dependent chromatic aberration (the p -offsets for Red, Green, and Blue channels) into print, they must be re-mapped to the subtractive CMYK color space.

- **Channel-Isolated Log Shifts:** Each CMYK plate receives its own unique p -offset to mimic the original optical dispersion. Crucially, the angular component θ for each plate is offset by traditional prepress screening angles (e.g., Black at 45°, Magenta at 75°) rewritten into log-polar angle differentials to completely eliminate structural moiré.

S7 p. 9 - prepress quantization, edge locking, and log-polar screening.

Appendix D - package contents and reproducibility

The package is designed as a self-contained technical corpus rather than a single narrative PDF. It contains normalized data, executable examples, source maps, visual diagrams, and the original documents.

Path	Purpose
Topological_Cross_Domain_Corpus_Report.pdf	Primary illustrated report.
Topological_Cross_Domain_Corpus_Report.html / .md	Editable and browser-readable report sources.
README.md	Package guide and quick start.
data/operator_lexicon.*	63-term cross-domain operator decoder.
data/claims_matrix.*	40-claim evidence and disposition matrix.
data/query_matrix.*	Query interface and failure-mode table.
data/normalization_rules.*	Motif-to-operator translation rules.
data/corpus_ontology.jsonld / .graphml	Machine-readable ontology and relationship graph.
specs/*.json, *.yaml, *.md	State/event schemas, API, compatibility, projection, and experiment specifications.
prototype/*.py	Restricted Equation World Zero implementation, demo, and tests.
diagrams/*.svg, *.png	Twelve report diagrams plus contact sheet.
source_maps/*	Per-document technical maps and key-page index.
source_extracts/*_FULL.txt	Full page-separated parsed source text.
source_documents/*.pdf	Original seven user-supplied source PDFs.
manifest.txt / SHA256SUMS.txt	File list and integrity hashes.

Reproduce the reference prototype

```
cd prototype
python demo.py
python -m unittest -v test_equation_world_zero.py
```

Evidence boundary

This report is a source-grounded synthesis and implementation scaffold. It is not external scientific validation, patent analysis, safety certification, medical advice, or proof that the proposed architecture outperforms existing systems.